

NONSOFC GROUPS EXIST

CLAUDE FABLE 5 AND GPT 5.6 SOL

ABSTRACT. We prove that nonsoc groups exist. The group $\text{EL}_4(L_{\mathbb{F}_2}(1, 2))$ of elementary matrices over the binary Leavitt algebra is infinite, finitely generated, has Kazhdan’s property (T) , and is not sofc; every invertible matrix entering the proof is written as an explicit product of elementary matrices, so no K -theory enters the headline result.

The analytic engine is a compression–centralizer criterion. If a Kazhdan group is generated by an infinite Kazhdan subgroup Γ together with finitely many elements that conjugate Γ into itself, then in a hypothetical sofc approximation the expander component decompositions supplied by Kun’s theorem for Γ and for the ambient group can be matched almost bijectively. The matching is conservative: a median normalization of component sizes is almost monotone along compressor arcs, exact conservation under permutations makes the estimate two-sided, and an elementary ℓ^1 co-area inequality on the ambient expanders pins the normalized sizes at their median. Localization on one matched component produces a sofc approximation of $K \times J$, where K is the compressed image of Γ and J is a commuting subgroup, in which the K -edges form an essential expander sequence; the Kun–Thom centralizer obstruction then forces J to be LEF. The Leavitt relations realize the scheme with J a copy of Thompson’s group V , which is infinite, simple, and finitely presented, hence not LEF.

Matrix self-similarity of $L = L_k(1, 2)$ spreads the conclusion. For every finite field k , the unit group $L^\times$, every $\text{GL}_r(L)$ with $r \geq 1$, and every $\text{EL}_r(L)$ with $r \geq 2$ are mutually isomorphic infinite finitely generated property (T) nonsoc groups, and the smallest matrix realization of the criterion occurs already in rank two. Finally, a finite multiplication-table construction shows that every finitely generated nonsoc group is a quotient of an infinite finitely presented nonsoc group, Kazhdan whenever the original group is. In particular, not every finitely presented discrete group is sofc.

CONTENTS

1. Introduction	2
Main results	3
Strategy	4
External inputs and dependency structure	5

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Organization	6
2. Sofic approximations, expanders, and LEF groups	6
2.1. Sofic approximations	6
2.2. Generator graphs and edit distance	7
2.3. Expansion and an ℓ^1 co-area estimate	7
2.4. LEF groups	8
2.5. The theorems of Kun and Kun–Thom	8
3. Conservative component matching: the toolkit	9
3.1. Expander refinement	9
3.2. Permutation conservation	9
3.3. Median pinning	10
3.4. Weighted selection	10
3.5. Localization by completion	11
4. The compression–centralizer criterion	11
4.1. Statement	11
4.2. The two component partitions	12
4.3. One-sided refinement along a compressor	13
4.4. Median normalization and conservation	14
4.5. Almost-bijective matching	15
4.6. Selection, localization, and the proof of Theorem 4.1	16
4.7. Two remarks on the mechanism	18
5. Leavitt algebras, elementary groups, and Thompson’s group V	19
5.1. Leaf sets and matrix self-similarity	19
5.2. Elementary groups over rings	20
5.3. Characteristic-two block factorizations	21
5.4. Thompson’s group V in a Leavitt corner	22
6. The compression scheme in adjacent ranks	23
6.1. The comb compressor	24
6.2. The involution z and generation	25
6.3. The commuting corner copy of V	26
6.4. The Leavitt-corner theorem	26
7. The explicit realization: no K -theory	27
8. Rank two and the full panorama	29
8.1. The two-by-two realization	29
8.2. K_1 vanishes and $GL = EL$	31
8.3. Finite fields: all ranks and the unit group	31
9. Finitely presented nonsofic covers	32
10. Concluding remarks	34
References	35

1. INTRODUCTION

A countable discrete group is *sofic* if every finite portion of its multiplication table admits arbitrarily accurate and asymptotically faithful models by

permutations of finite sets. The notion arose in Gromov's work on symbolic dynamics [10]; Weiss introduced the term and asked whether every countable group is sofic [21]. The class contains all amenable and all residually finite groups, and it is stable under a long list of standard constructions; see [7, 18]. Whether every group is sofic has remained a central open problem of modern group theory; a 2026 account still formulates the question in this form [8].

We answer the question in the negative.

Main results. Let k be a field and let

$$(1) \quad L = L_k(1, 2) = k\langle s_0, s_1, t_0, t_1 \mid t_i s_j = \delta_{ij}, s_0 t_0 + s_1 t_1 = 1 \rangle$$

be the binary Leavitt algebra [15]. For a unital ring A and $r \geq 2$, write

$$x_{ij}(a) = I_r + aE_{ij} \in \mathrm{GL}_r(A), \quad a \in A, \quad i \neq j,$$

and let $\mathrm{EL}_r(A) \leq \mathrm{GL}_r(A)$ be the subgroup generated by these elementary matrices.

Theorem A. *Let $R = L_{\mathbb{F}_2}(1, 2)$. For every $m \geq 3$, the group*

$$\mathrm{EL}_{m+1}(R)$$

is infinite, finitely generated, has property (T), and is not sofic. In particular $G = \mathrm{EL}_4(R)$ is not sofic, and not every countable discrete group is sofic. Every invertible matrix used in the proof is exhibited as an explicit product of elementary matrices; the proof uses no K -theory.

Theorem B. *Let k be a finite field and let $L = L_k(1, 2)$.*

- (i) *The unit group $L^\times$ is infinite, finitely generated, has property (T), and is not sofic.*
- (ii) *For every $r \geq 1$, every ordered complete binary leaf set $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ induces an explicit group isomorphism*

$$(2) \quad \Theta_{\mathcal{C}} : \mathrm{GL}_r(L) \xrightarrow{\cong} L^\times, \quad (a_{ij}) \mapsto \sum_{i,j=1}^r \alpha_i a_{ij} \alpha_j^*.$$

For $r \geq 2$ one has $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$. Hence all the groups $\mathrm{GL}_r(L)$, $r \geq 1$, and $\mathrm{EL}_r(L)$, $r \geq 2$, are mutually isomorphic, and each is an infinite finitely generated property (T) nonsocic group.

Theorem C. *Every finitely generated nonsocic group is a quotient of an infinite finitely presented nonsocic group. Every nonsocic group with property (T) is a quotient of an infinite finitely presented nonsocic group with property (T). In particular, there is an infinite finitely presented property (T) group H that is not sofic; H may be chosen to surject onto $L^\times$, and hence to have a quotient isomorphic to every group listed in Theorem B.*

Strategy. The proof has three independent layers.

The first layer is analytic.

Theorem D (Kazhdan compression–centralizer criterion). *Let G be a finitely generated property-(T) group, and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup. Suppose there is a finite set $Q \subseteq G$ with*

$$G = \langle \Gamma, Q \rangle \quad \text{and} \quad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

Fix $q_0 \in Q$, put $K = q_0\Gamma q_0^{-1}$, and let $J \leq \Gamma$ be finitely generated with

$$[K, J] = 1, \quad K \cap J = \{1\}.$$

If G is sofic, then J is LEF.

The point of departure is the tension between the two theorems that govern sofic approximations of Kazhdan groups. Kun’s theorem [13] proves Bowen’s conjecture: every sofic approximation of an infinite property-(T) group is essentially a vertex-disjoint union of uniformly expanding components. The Kun–Thom theorem [14] requires more: if the restriction of a sofic approximation of $\Lambda \times \Delta$ to a Kazhdan factor Λ is essentially *one* expander, then Δ is locally embeddable into finite groups (LEF). A disjoint union of equal-sized expanders is not enough, because the commuting factor may permute the components—this is exactly what happens in a product approximation.

Compression closes the gap. A hypothetical sofic approximation of G yields a Kun decomposition for Γ and another for G . Transporting the Γ -components by a compressor q gives one-sided inclusions into other Γ -components. Write $s(y)$ for the size of the Γ -component containing y ; on each ambient G -component, normalize by a vertex-weighted median m of the sizes and set

$$(3) \quad f(y) = \frac{s(y)}{s(y) + m}.$$

The one-sided inclusions make f almost monotone along compressor arcs. But a compressor acts by a permutation, so total increase equals total decrease *exactly*. The resulting small ℓ^1 edge variation, fed into an elementary co-area inequality on the ambient expanders, pins f at its median value $1/2$. Hence transported and target components have asymptotically equal sizes, and one-sided containment becomes almost-bijective matching. A weighted diagonal selection extracts a single matched component on which every multiplication, faithfulness, invariance, conjugacy, and graph-edit error is negligible. Restricting and completing the permutations yields a sofic approximation of $K \times J$ whose K -edges form an essential expander sequence, and the Kun–Thom obstruction forces J to be LEF. Theorem D is proved in Section 4 as a corollary of a local statement, Theorem 4.1, which isolates the exact graph-theoretic hypotheses and records precisely what property (T) is used for.

The second layer is algebraic, and it is where the Leavitt relations enter. Fix $m \geq 2$ and a unital ring A containing a binary Leavitt family. A left-comb binary partition with $m + 1$ leaves produces a compressor $u \in \mathrm{GL}_{m+1}(A)$ that conjugates the block subgroup $\Gamma \cong \mathrm{EL}_m(A)$ into the corner $a \mapsto s_0 a t_0$ of itself. A block involution z centralizes the compressed image K , so $v = zu$ is a second compressor with the same image, and $vu^{-1} = z$ generates every elementary matrix involving the last coordinate; thus $G = \mathrm{EL}_{m+1}(A) = \langle \Gamma, u, v \rangle$. The complementary corner $a \mapsto p_0 + s_1 a t_1$ carries a commuting copy of Thompson’s group V , which is infinite, simple, and finitely presented, hence not LEF. The corner copy of V is elementary *unconditionally*: V is perfect, and the Whitehead lemma places diagonal commutators of units in EL_2 . The only membership hypotheses of the general theorem therefore concern u and z . Over $\mathbb{F}_2$ they are discharged by explicit four-term elementary factorizations, which gives Theorem A with no K -theory; over every finite field they follow from $K_1(L) = 0$ and the GE property of purely infinite simple rings. Matrix self-similarity $M_r(L) \cong L$ then spreads nonsocicity and property (T) to the unit group and to all ranks, and a separate two-by-two realization shows that the scheme already closes in rank two.

The third layer is a finite multiplication-table construction. Every finitely generated nonsocic group G has a finite table fragment that admits no sufficiently accurate permutation model. Imposing precisely that finite table as relators gives a finitely presented group mapping onto G ; the map proves infinitude and preserves distinctness of the named elements, while the forbidden finite table proves nonsocicity directly. When G is Kazhdan, Shalom’s finite-presentation theorem first supplies a finitely presented Kazhdan cover, and imposing the same table inside its kernel produces a finitely presented Kazhdan nonsocic cover. No closure of socicity under quotients is used at any point.

External inputs and dependency structure. The proof is organized so that its trust surface is explicit.

Theorems A and D rest on exactly four external inputs: Kun’s expander decomposition theorem [13], the Kun–Thom centralizer obstruction [14], property (T) for elementary groups over finitely generated rings [9], and the classical facts that Thompson’s group V is infinite, simple, and finitely presented [11, 6, 5]. Everything else is self-contained. Every invertible matrix in the proof of Theorem A is factored explicitly into elementary matrices; the analytic sections use no spectral theory, the only inequality imported into the matching argument being the ℓ^1 co-area estimate of Lemma 2.5, proved in a few lines from the layer-cake formula.

Theorem B additionally uses $K_1(L) = 0$ and the GE property of purely infinite simple rings [1, 3, 2].

Theorem C is self-contained apart from Shalom’s finitely presented Kazhdan covers [19], used only for the Kazhdan refinement.

Organization. Section 2 fixes the sofic, expander, and LEF vocabulary and states the theorems of Kun and Kun–Thom in the forms used. Section 3 isolates the five elementary lemmas of the matching mechanism. Section 4 proves the local compression–centralizer theorem and deduces Theorem D. Section 5 develops Leavitt leaf calculus, matrix self-similarity, the Whitehead lemma, the characteristic-two block factorizations, and the corner realization of V . Section 6 presents the compression scheme in adjacent ranks over an arbitrary ring and proves the Leavitt-corner theorem. Section 7 discharges all memberships explicitly over $\mathbb{F}_2$ and proves Theorem A. Section 8 proves the two-by-two theorem, the equality $\text{GL}_r(L) = \text{EL}_r(L)$, and Theorem B. Section 9 proves Theorem C. Section 10 collects concluding remarks.

2. SOFIC APPROXIMATIONS, EXPANDERS, AND LEF GROUPS

2.1. Sofic approximations. For $\sigma, \tau \in \text{Sym}(Y)$, let

$$d_H(\sigma, \tau) = \frac{1}{|Y|} |\{y \in Y \mid \sigma(y) \neq \tau(y)\}|$$

denote the normalized Hamming distance.

Definition 2.1. Let G be a countable discrete group. A *sofic approximation* of G is a sequence of maps $\sigma_n : G \rightarrow \text{Sym}(Y_n)$, with $|Y_n| \rightarrow \infty$, such that

- (a) $d_H(\sigma_n(g)\sigma_n(h), \sigma_n(gh)) \rightarrow 0$ for all $g, h \in G$;
- (b) $d_H(\sigma_n(g), \text{id}) \rightarrow 0$ for all $g \in G \setminus \{1\}$.

The group G is *sofic* if it admits a sofic approximation.

The maps σ_n are not assumed to be homomorphisms. Property (b) self-improves: for a sofic group one may always pass to a sofic approximation in which each $\sigma_n(g)$, $g \neq 1$, moves *every* point of a set of density $1 - o(1)$, and it is standard that (a) and (b) force $d_H(\sigma_n(1), \text{id}) \rightarrow 0$. We will use the following exact normalization.

Lemma 2.2 (Finite normalization). *Let $\sigma_n : G \rightarrow \text{Sym}(Y_n)$ be a sofic approximation and let $F \subseteq G$ be a finite symmetric set. Then there is a sofic approximation $\sigma'_n : G \rightarrow \text{Sym}(Y_n)$ with $d_H(\sigma_n, \sigma'_n) \rightarrow 0$ pointwise, such that for all n :*

$$\sigma'_n(1) = \text{id}, \quad \sigma'_n(g^{-1}) = \sigma'_n(g)^{-1} \quad (g \in F).$$

Proof. Set $\sigma'_n(1) = \text{id}$, and note that $d_H(\sigma_n(1), \text{id}) \rightarrow 0$: applying (a) with $g = h = 1$ gives $d_H(\sigma_n(1)^2, \sigma_n(1)) \rightarrow 0$, and composing with the permutation $\sigma_n(1)^{-1}$ preserves d_H .

Choose one representative from each pair $\{g, g^{-1}\} \subseteq F \setminus \{1\}$ with $g \neq g^{-1}$; put $\sigma'_n(g) = \sigma_n(g)$ and $\sigma'_n(g^{-1}) = \sigma_n(g)^{-1}$. Then $d_H(\sigma_n(g^{-1}), \sigma_n(g)^{-1}) \rightarrow 0$, because $\sigma_n(g^{-1})\sigma_n(g)$ is d_H -close to $\sigma_n(1)$, hence to id , and d_H is invariant under composition with the permutation $\sigma_n(g)^{-1}$ on the right.

If $g = g^{-1} \in F \setminus \{1\}$ is an involution, let $A_n = \{y \mid \sigma_n(g)^2 y = y\}$. Since $\sigma_n(g)^2$ is d_H -close to $\sigma_n(g^2) = \sigma_n(1)$, hence to id , we have $|Y_n \setminus A_n| = o(|Y_n|)$. The set A_n is $\sigma_n(g)$ -invariant—if $\sigma_n(g)^2 y = y$, then applying $\sigma_n(g)$ gives

$\sigma_n(g)^2(\sigma_n(g)y) = \sigma_n(g)y$ —and $\sigma_n(g)$ restricts to an involution of A_n . Define $\sigma'_n(g)$ to be $\sigma_n(g)$ on A_n and the identity on $Y_n \setminus A_n$; this is an involution of Y_n differing from $\sigma_n(g)$ on $o(|Y_n|)$ points.

Outside $F \cup \{1\}$ put $\sigma'_n = \sigma_n$. Each $\sigma'_n(g)$ differs from $\sigma_n(g)$ on $o(|Y_n|)$ points, so properties (a) and (b) persist. $\square$

2.2. Generator graphs and edit distance. Let $T \subseteq G$ be a finite set and $\sigma : G \rightarrow \text{Sym}(Y)$ a map. The T -generator graph $X(Y, \sigma, T)$ is the graph on the vertex set Y whose edge multiset consists of the unordered pairs $\{y, \sigma(t)y\}$ with $y \in Y$, $t \in T$, and $\sigma(t)y \neq y$; we regard it as a graph of degree at most $2|T|$ after discarding multiplicities. An *arc* of $t \in T$ is an ordered pair $(y, \sigma(t)y)$.

For graphs X, X' on the same vertex set Y , the *edit distance* $\text{ed}(X, X')$ is the number of edges of the symmetric difference $E(X) \triangle E(X')$. A sequence of graphs X_n on Y_n is *edit-equivalent* to a sequence X'_n if $\text{ed}(X_n, X'_n) = o(|Y_n|)$.

For a finite graph X and $\emptyset \neq U \subseteq V(X)$, write ∂U for the set of edges with exactly one endpoint in U , and

$$h(X) = \min \left\{ \frac{|\partial U|}{|U|} \mid \emptyset \neq U \subseteq V(X), |U| \leq \frac{1}{2}|V(X)| \right\}$$

for the Cheeger constant. A sequence of bounded-degree graphs X_n with $|V(X_n)| \rightarrow \infty$ is an *expander sequence* if $\inf_n h(X_n) > 0$.

Definition 2.3. Let (X_n) be a sequence of graphs of uniformly bounded degree on vertex sets Y_n with $|Y_n| \rightarrow \infty$. An *expander decomposition* of (X_n) consists of graphs $\tilde{X}_n$ on Y_n , of uniformly bounded degree, with $\text{ed}(X_n, \tilde{X}_n) = o(|Y_n|)$, together with a constant $h > 0$ such that every connected component of every $\tilde{X}_n$ has Cheeger constant at least h .

Definition 2.4. A sequence of bounded-degree graphs (X_n) is an *essential expander sequence* if it is edit-equivalent to an expander sequence.

2.3. Expansion and an ℓ^1 co-area estimate. The single analytic inequality used in this paper is the following co-area estimate.

Lemma 2.5 (ℓ^1 co-area estimate). *Let X be a finite connected graph and $f : V(X) \rightarrow \mathbb{R}$. Let $c \in \mathbb{R}$ be such that*

$$|\{x \mid f(x) > c\}| \leq \frac{1}{2}|V(X)| \quad \text{and} \quad |\{x \mid f(x) < c\}| \leq \frac{1}{2}|V(X)|.$$

Then

$$\sum_{x \in V(X)} |f(x) - c| \leq \frac{1}{h(X)} \sum_{\{x, y\} \in E(X)} |f(x) - f(y)|.$$

Proof. Replacing f by $f - c$ we may assume $c = 0$. Write $f = f_+ - f_-$ with $f_{\pm} \geq 0$. For $t > 0$, the level set $U_t = \{x \mid f_+(x) > t\}$ satisfies $|U_t| \leq \frac{1}{2}|V(X)|$, so $|\partial U_t| \geq h(X)|U_t|$ whenever $U_t \neq \emptyset$. By the layer-cake formula,

$$\sum_x f_+(x) = \int_0^\infty |U_t| dt \leq \frac{1}{h(X)} \int_0^\infty |\partial U_t| dt \leq \frac{1}{h(X)} \sum_{\{x, y\} \in E(X)} |f_+(x) - f_+(y)|,$$

since an edge $\{x, y\}$ lies in ∂U_t exactly for t in an interval of length $|f_+(x) - f_+(y)|$. The same bound holds for f_- , and $|f_+(x) - f_+(y)| + |f_-(x) - f_-(y)| \leq |f(x) - f(y)|$ pointwise on edges, because f_+ and f_- have disjoint supports. Adding the two estimates gives the claim. $\square$

2.4. LEF groups. A group J is *LEF* (locally embeddable into finite groups) [20] if for every finite subset $F \subseteq J$ there are a finite group P and a map $\varphi : F \rightarrow P$ that is injective and satisfies $\varphi(g)\varphi(h) = \varphi(gh)$ whenever $g, h, gh \in F$. Every LEF group is sofic; finite and residually finite groups are LEF.

Lemma 2.6. *A finitely presented LEF group is residually finite.*

Proof. Let $J = \langle T \mid W \rangle$ with T, W finite, and let $g \in J \setminus \{1\}$. Choose a finite subset $F \subseteq J$ containing 1, g , the generators T , and all prefixes of all relators of W as words in T , and closed under the finitely many partial products needed to evaluate each relator step by step. An LEF embedding $\varphi : F \rightarrow P$ then yields an assignment $T \rightarrow P$, $t \mapsto \varphi(t)$, under which every relator of W evaluates to 1 in P : the evaluation proceeds through partial products lying in F , where φ is multiplicative. Hence $t \mapsto \varphi(t)$ extends to a homomorphism $J \rightarrow P$. This homomorphism sends g to $\varphi(g) \neq \varphi(1) = 1$ since φ is injective on F . Thus J is residually finite. $\square$

2.5. The theorems of Kun and Kun–Thom. We use two structure theorems for sofic approximations of Kazhdan groups.

Theorem 2.7 (Kun [13]). *Let Γ be an infinite finitely generated property-(T) group with a finite symmetric generating set S , and let $\sigma_n : \Gamma \rightarrow \text{Sym}(Y_n)$ be a sofic approximation. Then the S -generator graphs $X(Y_n, \sigma_n, S)$ admit an expander decomposition. Moreover, the modified graphs $\tilde{X}_n$ may be chosen to be of one fixed degree d , with the Cheeger constants of all components bounded below by a single constant $h > 0$; the constants d and h depend only on (Γ, S) .*

Kun’s theorem proves Bowen’s conjecture: sofic approximations of infinite Kazhdan groups are essentially disjoint unions of expanders. The companion result of Kun and Thom is an obstruction for centralizers.

Theorem 2.8 (Kun–Thom [14]). *Let Λ be an infinite finitely generated property-(T) group with finite symmetric generating set S_Λ , and let Δ be a finitely generated group. Suppose $\Lambda \times \Delta$ admits a sofic approximation $\pi_n : \Lambda \times \Delta \rightarrow \text{Sym}(D_n)$ such that the S_Λ -generator graphs $X(D_n, \pi_n|_\Lambda, S_\Lambda)$ form an essential expander sequence. Then Δ is LEF.*

Although the statement of their theorem is phrased for an expander sequence, the proof is carried out for an essentially expanding sequence; we record the form in which it is used. The heuristic content: a genuine expander is essentially connected, so the commuting factor Δ cannot shuffle components and is forced to act almost by automorphisms of finite structures, which yields LEF approximations.

3. CONSERVATIVE COMPONENT MATCHING: THE TOOLKIT

This section isolates five elementary lemmas about finite graphs, permutations, and sequences. They are stated once, in the exact quantitative forms used in Section 4; the only genuinely group-theoretic input of the paper appears there.

3.1. Expander refinement.

Lemma 3.1 (Expander refinement). *Let X be a finite graph, every connected component of which has Cheeger constant at least $h > 0$. Let $V(X) = A_1 \sqcup \cdots \sqcup A_r$ be a partition, and let e be the number of edges of X whose endpoints lie in different parts. Then there is an assignment $C \mapsto i(C)$ of a part to each component of X , with $|C \cap A_{i(C)}|$ maximal, such that*

$$\sum_C |C \setminus A_{i(C)}| \leq \frac{4e}{h}.$$

Proof. Every edge of X lies inside a component, so the crossing edges distribute among the components; let e_C be the number inside C , with $\sum_C e_C = e$. Fix a component C , let $a_1 \geq a_2 \geq \cdots \geq a_k$ be the sizes of its nonempty intersections with the parts, and let $A_{i(C)}$ realize a_1 . We claim

$$|C| - a_1 \leq \frac{4e_C}{h},$$

which proves the lemma upon summation. If a subset $W \subseteq C$ is a union of part-intersections with $|W| \leq \frac{1}{2}|C|$, then every edge of $\partial_{X[C]}W$ joins different parts, so $e_C \geq |\partial_{X[C]}W| \geq h|W|$.

If $a_1 \geq \frac{1}{2}|C|$, apply this to $W = C \setminus A_{i(C)}$, a union of part-intersections of size $|C| - a_1 \leq \frac{1}{2}|C|$; this gives $|C| - a_1 \leq e_C/h$.

If $a_1 < \frac{1}{2}|C|$, accumulate part-intersections in any order until the running total first reaches at least $\frac{1}{4}|C|$; the resulting union W satisfies $\frac{1}{4}|C| \leq |W| < \frac{1}{2}|C| + a_1 < \frac{3}{4}|C|$, so $\min(|W|, |C \setminus W|) \geq \frac{1}{4}|C|$, and whichever of $W, C \setminus W$ has size at most $\frac{1}{2}|C|$ is again a union of part-intersections. Applying the boundary bound to it gives $e_C \geq \frac{h}{4}|C| \geq \frac{h}{4}(|C| - a_1)$. $\square$

3.2. Permutation conservation.

Lemma 3.2 (Permutation conservation). *Let Y be a finite set, $\pi \in \text{Sym}(Y)$, and $f : Y \rightarrow \mathbb{R}$. Then*

$$\sum_{y \in Y} (f(y) - f(\pi y))_+ = \sum_{y \in Y} (f(\pi y) - f(y))_+ = \frac{1}{2} \sum_{y \in Y} |f(\pi y) - f(y)|.$$

Proof. Both $\sum_y f(y)$ and $\sum_y f(\pi y)$ equal the same number, so $\sum_y (f(y) - f(\pi y)) = 0$; split into positive and negative parts. $\square$

This exact identity is what converts a one-sided (monotonicity) estimate along the arcs of a permutation into a two-sided ℓ^1 estimate.

3.3. Median pinning.

Lemma 3.3 (Median pinning). *Let Z be a finite graph in which every connected component B satisfies $h(Z[B]) \geq h > 0$. Let $f : V(Z) \rightarrow \mathbb{R}$, and suppose that on every component at most half of the vertices satisfy $f > \frac{1}{2}$ and at most half satisfy $f < \frac{1}{2}$. Then*

$$\sum_{x \in V(Z)} \left| f(x) - \frac{1}{2} \right| \leq \frac{1}{h} \sum_{\{x,y\} \in E(Z)} |f(x) - f(y)|.$$

Proof. Every edge of Z lies within a component. Apply Lemma 2.5 with $c = \frac{1}{2}$ on each component and sum. $\square$

3.4. Weighted selection.

Lemma 3.4 (Weighted selection). *For each n , let $\{D_{n,i}\}_{i \in I_n}$ be disjoint nonempty subsets of a finite set Y_n , $|Y_n| = N_n \rightarrow \infty$, with $\sum_{i \in I_n} |D_{n,i}| \geq (1 - o(1))N_n$. Suppose that for every fixed M ,*

$$\sum_{i: |D_{n,i}| \leq M} |D_{n,i}| = o(N_n).$$

Let, for each $r \in \mathbb{N}$, nonnegative error weights $E_{n,i}^{(r)}$ be given with

$$\sum_{i \in I_n} E_{n,i}^{(r)} = o(N_n) \quad \text{for every fixed } r.$$

Then there exist indices $i(n) \in I_n$, for all n in a cofinite set, such that $|D_{n,i(n)}| \rightarrow \infty$ and, for every fixed r ,

$$E_{n,i(n)}^{(r)} = o(|D_{n,i(n)}|).$$

Proof. For $r \in \mathbb{N}$ put $\varepsilon_{n,r} = (\sum_i E_{n,i}^{(r)})/N_n$, so $\varepsilon_{n,r} \rightarrow 0$ as $n \rightarrow \infty$ for each fixed r . Choose $R_n \rightarrow \infty$ growing so slowly that $\delta_n := \max_{r \leq R_n} \varepsilon_{n,r} \rightarrow 0$ and additionally, writing ρ_n for the total mass of the sets with $|D_{n,i}| \leq \delta_n^{-1/2}$ plus the deficiency $N_n - \sum_i |D_{n,i}|$, that $\rho_n = o(N_n)$; this is possible by the small-mass hypothesis applied along a diagonal. Call $i \in I_n$ *bad* for r if $E_{n,i}^{(r)} > \delta_n^{1/2} |D_{n,i}| 2^{-r}$. The total mass of sets bad for $r \leq R_n$ is at most

$$\sum_{r \leq R_n} \frac{2^r}{\delta_n^{1/2}} \sum_i E_{n,i}^{(r)} \leq \frac{N_n}{\delta_n^{1/2}} \sum_{r \leq R_n} 2^r \varepsilon_{n,r},$$

which, after shrinking R_n further if necessary so that $\sum_{r \leq R_n} 2^r \varepsilon_{n,r} \leq \delta_n^{3/4}$, is at most $\delta_n^{1/4} N_n = o(N_n)$. Since the good sets (not bad for any $r \leq R_n$, of size larger than $\delta_n^{-1/2}$) then carry mass $(1 - o(1))N_n > 0$ for large n , pick $i(n)$ among them. Then $|D_{n,i(n)}| > \delta_n^{-1/2} \rightarrow \infty$, and for each fixed r , once $R_n \geq r$,

$$E_{n,i(n)}^{(r)} \leq \delta_n^{1/2} 2^{-r} |D_{n,i(n)}| = o(|D_{n,i(n)}|). \quad \square$$

3.5. Localization by completion.

Lemma 3.5 (Localization by completion). *Let H be a countable group and $\sigma_n : H \rightarrow \text{Sym}(Y_n)$ maps. Suppose $D_n \subseteq Y_n$ are subsets with $|D_n| \rightarrow \infty$ such that, for every fixed $g, h \in H$:*

- (i) $|\{y \in D_n \mid \sigma_n(g)y \notin D_n\}| = o(|D_n|)$;
- (ii) $|\{y \in D_n \mid \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y\}| = o(|D_n|)$;
- (iii) $|\{y \in D_n \mid \sigma_n(g)y = y\}| = o(|D_n|)$ whenever $g \neq 1$.

For $g \in H$ define $\pi_n(g) \in \text{Sym}(D_n)$ by letting $\pi_n(g)$ agree with $\sigma_n(g)$ on $\{y \in D_n \mid \sigma_n(g)y \in D_n, \sigma_n(g)y \neq \sigma_n(g)y' \forall y' \neq y \in D_n\}$ and extending the resulting partial injection of D_n into itself to a permutation of D_n arbitrarily. Then $\pi_n : H \rightarrow \text{Sym}(D_n)$ is a sofic approximation of H . Moreover, if $g, g^{-1} \in H$ satisfy $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$, the completions may be chosen with $\pi_n(g^{-1}) = \pi_n(g)^{-1}$, and then

$$|\{y \in D_n \mid \pi_n(g)y \neq \sigma_n(g)y\}| = o(|D_n|).$$

Proof. Since $\sigma_n(g)$ is injective on Y_n , the described set is all of the set of points of D_n sent into D_n by $\sigma_n(g)$, whose complement in D_n has size $o(|D_n|)$ by (i); the partial map is an injection, and any extension to a permutation of D_n differs from $\sigma_n(g)$ on $o(|D_n|)$ points of D_n . Then (ii) transfers: for fixed g, h , the points of D_n where $\pi_n(g)\pi_n(h)y \neq \pi_n(gh)y$ are contained in the union of the $o(|D_n|)$ -sized exceptional sets for g, h, gh (points where $\pi \neq \sigma$, including the $\sigma_n(h)$ -preimage of the exceptional set of g) and the set in (ii); similarly (iii) transfers faithfulness. When $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$, the two partial injections attached to g and to g^{-1} are mutually inverse, so a completion of one can be transported to the other. $\square$

4. THE COMPRESSION-CENTRALIZER CRITERION

This section proves the analytic core of the paper. We first state a local version that isolates the exact graph-theoretic hypotheses; property (T) of the ambient group is a sufficient condition for the second hypothesis, and the local statement records precisely what it is used for.

4.1. Statement.

Theorem 4.1 (Local compression-centralizer theorem). *Let G be a finitely generated group and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup with finite symmetric generating set S . Let $Q \subseteq G$ be a finite set with*

$$(4) \quad G = \langle \Gamma, Q \rangle \quad \text{and} \quad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

Fix $q_0 \in Q$, put $K = q_0\Gamma q_0^{-1}$, and let $J \leq \Gamma$ be finitely generated with

$$(5) \quad [K, J] = 1, \quad K \cap J = \{1\}.$$

Put $A = S \cup Q \cup Q^{-1}$. Suppose G admits a sofic approximation $\sigma_n : G \rightarrow \text{Sym}(Y_n)$ such that

- (i) the S -generator graphs $X(Y_n, \sigma_n, S)$ admit an expander decomposition, and
- (ii) the A -generator graphs $X(Y_n, \sigma_n, A)$ admit an expander decomposition.

Then J is LEF.

Corollary 4.2 (Kazhdan compression–centralizer criterion; Theorem D). *In the setting of (4) and (5), suppose in addition that G and Γ have property (T). If G is sofic, then J is LEF. Equivalently: if J is not LEF, then G is not sofic.*

Proof. Suppose G is sofic and fix a sofic approximation σ_n . By (4) the finite symmetric set $A = S \cup Q \cup Q^{-1}$ generates G . The restriction $\sigma_n|_\Gamma$ is a sofic approximation of the infinite Kazhdan group Γ , so Theorem 2.7 yields hypothesis (i); applying Theorem 2.7 to G itself with the generating set A yields hypothesis (ii). Theorem 4.1 concludes. $\square$

Remark 4.3. Only the componentwise ℓ^1 median inequality of Lemma 3.3 is extracted from hypothesis (ii); that hypothesis may accordingly be weakened to any decomposition of the A -generator graphs, up to $o(|Y_n|)$ edge edits, into components satisfying a uniform ℓ^1 Poincaré inequality around medians. This separates the approximation-theoretic input, which property (T) supplies through Kun’s theorem, from the compression mechanism, which is combinatorial.

The remainder of the section proves Theorem 4.1. Write $N_n = |Y_n|$. All $o(N_n)$ estimates are uniform over the fixed finite data (S, Q, q_0) and the decomposition constants; the set Q itself is never symmetrized inside the compression steps, since the compression hypothesis $q\Gamma q^{-1} \leq \Gamma$ is assumed only for $q \in Q$.

4.2. The two component partitions. Put $S_K = q_0 S q_0^{-1}$, a finite symmetric generating set of K . By Lemma 2.2 we may and do assume the exact identities

$$\sigma_n(1) = \text{id}, \quad \sigma_n(g^{-1}) = \sigma_n(g)^{-1} \quad (g \in A \cup S_K),$$

after a pointwise $o(N_n)$ modification; each relevant generator graph changes by $o(N_n)$ edges, so hypotheses (i) and (ii) persist.

By hypothesis (i), fix an expander decomposition of the S -generator graphs: graphs $\tilde{X}_n$ on Y_n of degree at most d_Γ , with

$$\text{ed}(X(Y_n, \sigma_n, S), \tilde{X}_n) = o(N_n),$$

such that every component of $\tilde{X}_n$ has Cheeger constant at least $h_\Gamma > 0$. Let $\mathcal{P}_n$ denote the partition of Y_n into the components of $\tilde{X}_n$, and write $P_n(y)$ for the block containing y . We refer to the blocks of $\mathcal{P}_n$ as Γ -components.

By hypothesis (ii), fix likewise an expander decomposition $\tilde{Z}_n$ of the A -generator graphs, with all components of Cheeger constant at least $h_G > 0$ and uniformly bounded degree; let $\mathcal{B}_n$ be the partition of Y_n into the components of $\tilde{Z}_n$, with blocks $B_n(y)$, the G -components.

Both partitions are almost invariant in the following sense. For every $s \in S$,

$$(6) \quad |\{y \in Y_n \mid P_n(\sigma_n(s)y) \neq P_n(y)\}| = o(N_n),$$

because apart from the $o(N_n)$ edge edits, every arc $(y, \sigma_n(s)y)$ is an edge of $\tilde{X}_n$ or a loop, and edges of $\tilde{X}_n$ do not cross $\mathcal{P}_n$. For every $a \in A$,

$$(7) \quad |\{y \in Y_n \mid B_n(\sigma_n(a)y) \neq B_n(y)\}| = o(N_n),$$

for the same reason applied to $\tilde{Z}_n$. Since $S \subseteq A$, every Γ -component is, up to a total error of $o(N_n)$ vertices over all of Y_n , contained in a single G -component: applying Lemma 3.1 to $\tilde{X}_n$ with the partition $\mathcal{B}_n$, whose crossing edges number $o(N_n)$ by (7) and the edit bounds, and discarding the leakage, we may—after moving $o(N_n)$ vertices into singleton blocks of both partitions, which affects no estimate below—assume that

$$(8) \quad \text{every block of } \mathcal{P}_n \text{ is contained in a block of } \mathcal{B}_n.$$

4.3. One-sided refinement along a compressor. Fix $q \in Q$. The group identity $qs = (qsq^{-1})q$ and approximate multiplicativity give, for each $s \in S$, the *conjugacy defect* bound

$$(9) \quad |\{y \in Y_n \mid \sigma_n(q)\sigma_n(s)y \neq \sigma_n(qsq^{-1})\sigma_n(q)y\}| = o(N_n).$$

Here qsq^{-1} belongs to the fixed finite set $qSq^{-1} \subseteq \Gamma$.

Transport the graphs $\tilde{X}_n$ by the permutation $\sigma_n(q)$: the transported graph $\sigma_n(q)\tilde{X}_n$ has vertex set Y_n , edge set $\{\{\sigma_n(q)x, \sigma_n(q)y\} \mid \{x, y\} \in E(\tilde{X}_n)\}$, components $\{\sigma_n(q)C \mid C \in \mathcal{P}_n\}$, and the same component Cheeger constants $\geq h_\Gamma$ and degree bound d_Γ . By (9) and the edit bounds, apart from $o(N_n)$ edge occurrences the edges of the transported graph are of the form $\{\sigma_n(q)y, \sigma_n(\gamma)\sigma_n(q)y\}$ with $\gamma \in qSq^{-1} \subseteq \Gamma$; each such γ is a word in S , so by (6), applied to the letters of the finitely many fixed words, all but $o(N_n)$ of the transported edges lie inside blocks of $\mathcal{P}_n$.

Apply Lemma 3.1 to the transported graph with the partition $\mathcal{P}_n$. This produces a map

$$\Phi_{q,n} : \mathcal{P}_n \rightarrow \mathcal{P}_n, \quad \Phi_{q,n}(C) = \text{a block of } \mathcal{P}_n \text{ meeting } \sigma_n(q)C \text{ in maximal size,}$$

with total leakage

$$(10) \quad \sum_{C \in \mathcal{P}_n} e_{q,n}(C) = o(N_n), \quad \text{where} \quad e_{q,n}(C) := |\sigma_n(q)C \setminus \Phi_{q,n}(C)|.$$

Thus each transported Γ -component is almost contained in a single target Γ -component. This containment is a priori one-sided: nothing yet prevents a large target from absorbing a small transported component.

4.4. Median normalization and conservation. For $y \in Y_n$, let $s_n(y) = |P_n(y)|$ be the size of its Γ -component. For a G -component $B \in \mathcal{B}_n$, let $m_n(B) \geq 1$ be a *vertex-weighted median* of s_n on B : a value such that

$$|\{y \in B \mid s_n(y) > m_n(B)\}| \leq \frac{1}{2}|B| \quad \text{and} \quad |\{y \in B \mid s_n(y) < m_n(B)\}| \leq \frac{1}{2}|B|.$$

Define the *median normalization*

$$(11) \quad f_n(y) = \frac{s_n(y)}{s_n(y) + m_n(B_n(y))} \in (0, 1).$$

Since $t \mapsto t/(t+m)$ is strictly increasing, on every G -component B at most half the vertices satisfy $f_n > \frac{1}{2}$ and at most half satisfy $f_n < \frac{1}{2}$; that is, $\frac{1}{2}$ is a median of f_n on every component of $\tilde{Z}_n$.

Variation along S . For $s \in S$,

$$(12) \quad \sum_{y \in Y_n} |f_n(\sigma_n(s)y) - f_n(y)| = o(N_n),$$

since the summand vanishes unless $P_n(\sigma_n(s)y) \neq P_n(y)$ or $B_n(\sigma_n(s)y) \neq B_n(y)$, a set of size $o(N_n)$ by (6) and (7), and $0 < f_n < 1$.

One-sided variation along $q \in Q$. Call $y \in Y_n$ a *good point* for q if

$$(13) \quad \sigma_n(q)y \in \Phi_{q,n}(P_n(y)) \quad \text{and} \quad B_n(\sigma_n(q)y) = B_n(y).$$

Let y be good, $C = P_n(y)$, $D = \Phi_{q,n}(C)$, and $e = e_{q,n}(C)$; then

$$s_n(\sigma_n(q)y) = |D| \geq |\sigma_n(q)C \cap D| = |C| - e = s_n(y) - e.$$

Since both y and $\sigma_n(q)y$ lie in the same G -component B , they are normalized by the same median $m = m_n(B) \geq 1$. For $a \geq (1-\delta)a' \geq 0$ and $m \geq 1$,

$$(14) \quad \frac{a'}{a' + m} - \frac{a}{a + m} \leq \frac{a'}{a' + m} - \frac{(1-\delta)a'}{(1-\delta)a' + m} \leq \delta,$$

where the first inequality is monotonicity of $t \mapsto t/(t+m)$ and the second is a direct computation: the difference equals $\delta a' m / ((a' + m)((1-\delta)a' + m)) \leq \delta$. Applying (14) with $a' = |C|$, $a = |D| \geq |C| - e$, $\delta = e/|C|$ gives, for every good point y ,

$$(f_n(y) - f_n(\sigma_n(q)y))_+ \leq \frac{e_{q,n}(P_n(y))}{|P_n(y)|}.$$

Summing over Y_n —good points of a block C contribute at most $e_{q,n}(C)$ in total, points with $\sigma_n(q)y \notin \Phi_{q,n}(P_n(y))$ number $e_{q,n}(C)$ per block and contribute at most 1 each, and points whose arc crosses $\mathcal{B}_n$ contribute at most 1 each—we obtain

$$(15) \quad \begin{aligned} \sum_{y \in Y_n} (f_n(y) - f_n(\sigma_n(q)y))_+ &\leq 2 \sum_{C \in \mathcal{P}_n} e_{q,n}(C) + |\{y \mid B_n(\sigma_n(q)y) \neq B_n(y)\}| \\ &= o(N_n) \end{aligned}$$

by (10) and (7). The compressed image can only grow—but growth is globally impossible:

Conservation. Since $\sigma_n(q)$ is a permutation of Y_n , Lemma 3.2 upgrades (15) to the two-sided estimate

$$(16) \quad \sum_{y \in Y_n} |f_n(\sigma_n(q)y) - f_n(y)| = 2 \sum_{y \in Y_n} (f_n(y) - f_n(\sigma_n(q)y))_+ = o(N_n) \quad (q \in Q),$$

and the change of variables $y = \sigma_n(q)x$, legitimate because $\sigma_n(q^{-1}) = \sigma_n(q)^{-1}$, extends (16) to all $q \in Q \cup Q^{-1}$.

Pinning. Every edge of the A -generator graph $X(Y_n, \sigma_n, A)$ arises from an arc $(y, \sigma_n(a)y)$ with $a \in A = S \cup Q \cup Q^{-1}$, and each edge arises from at most $2|A|$ arcs. Hence (12) and (16) give

$$\sum_{\{x,y\} \in E(X(Y_n, \sigma_n, A))} |f_n(x) - f_n(y)| = o(N_n).$$

Since $0 < f_n < 1$, editing $o(N_n)$ edges changes this sum by at most $o(N_n)$; therefore the same bound holds over $E(\tilde{Z}_n)$. As $\frac{1}{2}$ is a median of f_n on every component of $\tilde{Z}_n$ and all component Cheeger constants are at least h_G , Lemma 3.3 yields the concentration estimate

$$(17) \quad \sum_{y \in Y_n} \left| f_n(y) - \frac{1}{2} \right| = o(N_n).$$

In words: after median normalization, almost every Γ -component has size asymptotically equal to the median size on its G -component, because $s/(s+m) \rightarrow \frac{1}{2}$ exactly when $s/m \rightarrow 1$.

4.5. Almost-bijective matching.

Proposition 4.4 (Matching). *Fix $q \in Q$. There are subcollections $\mathcal{G}_{q,n} \subseteq \mathcal{P}_n$ such that:*

- (i) $\sum_{C \in \mathcal{P}_n \setminus \mathcal{G}_{q,n}} |C| = o(N_n)$;
 - (ii) for every $C \in \mathcal{G}_{q,n}$, with $D = \Phi_{q,n}(C)$,
- $$(18) \quad |\sigma_n(q)C \triangle D| = o(1) \cdot |C| \quad \text{uniformly over } C \in \mathcal{G}_{q,n};$$
- (iii) for n large, $\Phi_{q,n}$ is injective on $\mathcal{G}_{q,n}$;
 - (iv) $\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1 - o(1)) N_n$.

Proof. Choose $\eta_n \downarrow 0$ so slowly that the three global errors—in (7) for the element q , in (10), and in (17)—are all $o(\eta_n^2 N_n)$. Let $W_n \subseteq Y_n$ be the union of the four sets

$$\begin{aligned} & \left\{ y \mid \left| f_n(y) - \frac{1}{2} \right| > \eta_n \right\}, \quad \sigma_n(q)^{-1} \left\{ y \mid \left| f_n(y) - \frac{1}{2} \right| > \eta_n \right\}, \\ & \{ y \mid B_n(\sigma_n(q)y) \neq B_n(y) \}, \quad \{ y \mid \sigma_n(q)y \notin \Phi_{q,n}(P_n(y)) \}. \end{aligned}$$

Markov's inequality applied to (17) bounds the first set by $\eta_n^{-1} \cdot o(\eta_n^2 N_n) = o(\eta_n N_n)$; the second is a bijective preimage of the first; the last two have sizes $o(\eta_n^2 N_n)$ by the choice of η_n . Hence $|W_n| = o(\eta_n N_n)$.

Discard from $\mathcal{P}_n$ every block C with $e_{q,n}(C) > \eta_n |C|$ or $|C \cap W_n| > \eta_n |C|$, and call the remaining collection $\mathcal{G}_{q,n}$. The discarded mass is at most $\eta_n^{-1} \sum_C e_{q,n}(C) + \eta_n^{-1} |W_n| = o(N_n)$, proving (i).

Let $C \in \mathcal{G}_{q,n}$ and pick $y \in C \setminus W_n$, possible since $\eta_n < 1$. Put $D = \Phi_{q,n}(C)$ and $B = B_n(y) = B_n(\sigma_n(q)y)$, and write $m = m_n(B)$. Since $y \notin W_n$, we have $\sigma_n(q)y \in D$, so $s_n(\sigma_n(q)y) = |D|$, and

$$\left| \frac{|C|}{|C| + m} - \frac{1}{2} \right| \leq \eta_n, \quad \left| \frac{|D|}{|D| + m} - \frac{1}{2} \right| \leq \eta_n.$$

Solving $\left| \frac{a}{a+m} - \frac{1}{2} \right| \leq \eta$ gives $\frac{1-2\eta}{1+2\eta} m \leq a \leq \frac{1+2\eta}{1-2\eta} m$; applying this to $a = |C|$ and to $a = |D|$ and dividing,

$$(19) \quad |D| = (1 + O(\eta_n)) |C| \quad \text{uniformly over } C \in \mathcal{G}_{q,n}.$$

With $e = e_{q,n}(C) \leq \eta_n |C|$,

$$\begin{aligned} |\sigma_n(q)C \triangle D| &= |\sigma_n(q)C \setminus D| + |D \setminus \sigma_n(q)C| \\ &\leq e + (|D| - |C| + e) \leq 2e + ||D| - |C|| = O(\eta_n)|C|, \end{aligned}$$

proving (ii).

For (iii): if $C \neq C'$ in $\mathcal{G}_{q,n}$ had $\Phi_{q,n}(C) = \Phi_{q,n}(C') = D$, the disjoint sets $\sigma_n(q)C \cap D$ and $\sigma_n(q)C' \cap D$ would, by (18) and (19), each occupy a fraction $1 - o(1)$ of D , uniformly—impossible for n large.

For (iv): by (i), (ii), and (19),

$$\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1+o(1)) \sum_{C \in \mathcal{G}_{q,n}} |C| = (1+o(1))(1-o(1))N_n = (1-o(1))N_n.$$

□

Thus one-sided refinement, conservation, and pinning together force the compressor to permute the Γ -components almost bijectively, with asymptotically equal sizes. One more observation is needed before selecting a single matched pair.

Lemma 4.5. *For every fixed M , $\sum \{|C| : C \in \mathcal{P}_n, |C| \leq M\} = o(N_n)$.*

Proof. Let $g_1, \dots, g_{M+1} \in \Gamma$ be distinct elements; they exist since Γ is infinite. By soficity, for all but $o(N_n)$ points y the points $\sigma_n(g_1)y, \dots, \sigma_n(g_{M+1})y$ are pairwise distinct; writing each g_i as a word in S and using (6) along its letters, for all but $o(N_n)$ points y all these points lie in $P_n(y)$. A point y satisfying both conditions has $|P_n(y)| \geq M+1$; hence the mass of blocks of size at most M is $o(N_n)$. □

4.6. Selection, localization, and the proof of Theorem 4.1.

Proof of Theorem 4.1. Apply Proposition 4.4 with $q = q_0$. For n large this gives a collection of *matched pairs*

$$(20) \quad \{(C, D) : C \in \mathcal{G}_{q_0,n}, D = \Phi_{q_0,n}(C)\},$$

with pairwise distinct sources, pairwise distinct targets, target mass $(1 - o(1))N_n$, and $|\sigma_n(q_0)C \triangle D| = o(1)|C|$ uniformly.

Fix an exhaustion $L_1 \subseteq L_2 \subseteq \dots$ of the countable group $KJ \leq \Gamma$ by finite subsets with $\bigcup_r L_r = KJ$, each L_r containing 1, the finite set S_K , and closed under inverses, and such that $L_r \cdot L_r \subseteq L_{r+1}$. To each matched pair (C, D) and each r attach the error weight $E^{(r)}(C, D)$, the sum of:

- (a) $|\sigma_n(q_0)C \triangle D|$;
- (b) the number of edges of $\tilde{X}_n$ with an endpoint in C or D that are edited relative to the S -generator graph, plus the number of edges of the transported graph $\sigma_n(q_0)\tilde{X}_n$ with an endpoint in $D \cup \sigma_n(q_0)C$ that are edited relative to the transported S -generator graph;
- (c) $|\{y \in C \mid \sigma_n(q_0)\sigma_n(s)y \neq \sigma_n(q_0sq_0^{-1})\sigma_n(q_0)y \text{ for some } s \in S\}|$;
- (d) $|\{y \in D \mid \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y \text{ for some } g, h \in L_r\}|$;
- (e) $|\{y \in D \mid \sigma_n(g)y = y \text{ for some } g \in L_r \setminus \{1\}\}|$;
- (f) $|\{y \in D \mid P_n(\sigma_n(g)y) \neq P_n(y) \text{ for some } g \in L_r\}|$.

For every fixed r , the sum of $E^{(r)}$ over the matched pairs is $o(N_n)$: (a) by Proposition 4.4(ii) and (i); (b) because the sources are pairwise disjoint and the targets are pairwise disjoint, so each edited edge is counted a bounded number of times, and the global edit counts are $o(N_n)$; (c) by (9) over the finite set S , the sources being disjoint; (d) and (e) by soficity of σ_n over the finite set $L_r \cup L_r^2$, the targets being disjoint; (f) by (6) applied along the letters of a fixed word in S for each $g \in L_r \subseteq \Gamma$, the targets being disjoint.

The targets D are pairwise disjoint blocks of $\mathcal{P}_n$ with total mass $(1 - o(1))N_n$, and by Lemma 4.5 the total mass of targets of size at most M is $o(N_n)$ for every fixed M . Lemma 3.4 therefore produces matched pairs (C_n, D_n) with

$$(21) \quad |D_n| \rightarrow \infty, \quad E^{(r)}(C_n, D_n) = o(|D_n|) \quad \text{for every fixed } r.$$

Put $U_n = \sigma_n(q_0)C_n$, so $|U_n \triangle D_n| = o(|D_n|)$ and $|U_n| = |C_n| = (1 + o(1))|D_n|$. Define

$$(22) \quad D'_n = U_n,$$

a set of size $|C_n| \rightarrow \infty$ that agrees with D_n up to $o(|D_n|)$ points; every error weight in (21) transfers from D_n to D'_n at the cost of $o(|D'_n|)$.

Localization. Consider the countable group $H = KJ \leq \Gamma$. By (5), K and J commute and intersect trivially, so

$$H = KJ \cong K \times J.$$

For every fixed $g \in H$, the transferred error (f) shows $|\{y \in D'_n \mid \sigma_n(g)y \notin D_n\}| = o(|D'_n|)$, hence $|\{y \in D'_n \mid \sigma_n(g)y \notin D'_n\}| = o(|D'_n|)$; the transferred errors (d) and (e) give multiplicativity and faithfulness on D'_n for every fixed pair $g, h \in H$, by choosing r with $g, h \in L_r$. Lemma 3.5, with completions on the symmetric set S_K chosen compatibly with the exact inverse identities of Section 4.2, yields a sofic approximation

$$\pi_n : K \times J \longrightarrow \text{Sym}(D'_n)$$

with $|\{y \in D'_n \mid \pi_n(g)y \neq \sigma_n(g)y\}| = o(|D'_n|)$ for every fixed g , and $\pi_n(g^{-1}) = \pi_n(g)^{-1}$ for $g \in S_K$.

The K -edges expand. Restrict $\tilde{X}_n$ to C_n and transport by $\sigma_n(q_0)$: the resulting graph Ξ_n on $D'_n = \sigma_n(q_0)C_n$ is connected with Cheeger constant at least h_Γ and degree at most d_Γ , because C_n is a component of $\tilde{X}_n$. We claim that Ξ_n and the S_K -generator graph $X(D'_n, \pi_n, S_K)$ differ by $o(|D'_n|)$ edges. Indeed, an edge of either graph that fails to be an edge of the other must involve one of the following, each affecting $o(|D'_n|)$ edge occurrences by (21):

- (1) an edge of $\tilde{X}_n[C_n]$ edited relative to the S -generator graph on C_n , or conversely (error (b));
- (2) an arc $(y, \sigma_n(s)y)$, $y \in C_n$, whose $\sigma_n(q_0)$ -transport disagrees with the $\sigma_n(q_0 s q_0^{-1})$ -arc at $\sigma_n(q_0)y$ (error (c));
- (3) a point of D'_n where $\pi_n(g) \neq \sigma_n(g)$ for some $g \in S_K$ (localization defect);
- (4) an arc of some $g \in S_K$ leaving D'_n (transferred error (f) and the symmetric difference (a)).

Hence the S_K -generator graphs of $\pi_n|_K$ form an essential expander sequence in the sense of Definition 2.4.

Conclusion. The group $K = q_0 \Gamma q_0^{-1} \cong \Gamma$ is an infinite finitely generated property-(T) group with symmetric generating set S_K , the group J is finitely generated, and π_n is a sofic approximation of $K \times J$ whose S_K -generator graphs form an essential expander sequence. Theorem 2.8 applies and shows that J is LEF. $\square$

4.7. Two remarks on the mechanism.

Remark 4.6 (The product obstruction). The compression hypothesis cannot be deleted from Theorem D. For a direct product $\Lambda \times \Delta$ of an infinite Kazhdan group with a residually finite Kazhdan group, product sofic approximations exist whenever Λ is sofic, and in them the Δ -factor genuinely permutes the Λ -components: the Λ -component partition has all blocks of equal size, and no single component is almost invariant under the second factor. The role of the compressors $q \in Q$, through (4), is to generate the whole ambient group while conjugating Γ into itself, so that the ambient expansion of hypothesis (ii) couples the component structure of Γ to itself across all of G and forbids exactly this shuffling.

Remark 4.7 (The hypothesis $J \leq \Gamma$ is essential). The subgroup J is required to lie in Γ , and this is used twice: through (6) in error (f), which keeps the J -arcs inside the matched component, and through the localization step, which restricts σ_n on elements of $KJ \leq \Gamma$. The hypothesis cannot be dropped in favor of $J \leq G$. Indeed, groups satisfying all the remaining hypotheses with a non-LEF subgroup $J \leq G$ commuting with K exist in abundance among sofic groups—already $J = \text{BS}(2, 3) = \langle a, t \mid ta^2t^{-1} = a^3 \rangle$ [4] is a finitely presented, residually solvable, hence sofic group that is non-Hopfian, therefore not residually finite, therefore not LEF by Lemma 2.6. The

criterion is calibrated exactly so that the commuting subgroup rides inside the compressed expander, and this is what the Leavitt corner construction of Sections 6–7 arranges algebraically.

5. LEAVITT ALGEBRAS, ELEMENTARY GROUPS, AND THOMPSON'S GROUP V

Throughout, k is a field and $L = L_k(1, 2)$ is the binary Leavitt algebra (1), with generators s_0, s_1, t_0, t_1 and relations $t_i s_j = \delta_{ij}$, $s_0 t_0 + s_1 t_1 = 1$. Write

$$p_0 = s_0 t_0, \quad p_1 = s_1 t_1,$$

a pair of orthogonal idempotents summing to 1: $p_i p_j = \delta_{ij} p_i$, since $p_0 p_1 = s_0(t_0 s_1)t_1 = 0$ and $p_i^2 = s_i(t_i s_i)t_i = p_i$. For a binary word $w = w_1 \cdots w_\ell \in \{0, 1\}^*$, put $s_w = s_{w_1} \cdots s_{w_\ell}$ and $s_w^* = t_{w_\ell} \cdots t_{w_1}$, so that $s_w^* s_{w'} = \delta_{w, w'}$ whenever $|w| = |w'|$, and more generally $s_w^* s_{w'}$ vanishes unless one of w, w' is a prefix of the other. The algebra L is infinite-dimensional over k : the elements $s_{0^i s_1}$, $i \geq 0$, are killed on the left by mutually orthogonal projections and are linearly independent.

5.1. Leaf sets and matrix self-similarity. A finite set of binary words $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ (we use the same symbols for the words and for the corresponding elements $s_{\alpha_i} \in L$) is a *complete leaf set* if the words are the leaves of a finite full rooted binary tree; equivalently, if they are pairwise prefix-incomparable and the cylinders they define partition the boundary $\{0, 1\}^{\mathbb{N}}$.

Lemma 5.1 (Leaf identities). *Let $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ be a complete leaf set. Then in L :*

$$\alpha_i^* \alpha_j = \delta_{ij} \quad \text{and} \quad \sum_{i=1}^r \alpha_i \alpha_i^* = 1.$$

Proof. Prefix-incomparability and the relations $t_i s_j = \delta_{ij}$ give the first identity. The second holds by induction on the tree: for the trivial tree it reads $1 = 1$; replacing a leaf α by its two children $\alpha 0, \alpha 1$ changes the sum by

$$s_{\alpha 0} s_{\alpha 0}^* + s_{\alpha 1} s_{\alpha 1}^* - s_{\alpha} s_{\alpha}^* = s_{\alpha} (s_0 t_0 + s_1 t_1 - 1) s_{\alpha}^* = 0. \quad \square$$

Proposition 5.2 (Matrix self-similarity). *Let $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ be an ordered complete leaf set. Then*

$$\Theta_{\mathcal{C}} : M_r(L) \longrightarrow L, \quad (a_{ij}) \longmapsto \sum_{i,j=1}^r \alpha_i a_{ij} \alpha_j^*,$$

is an isomorphism of unital k -algebras, with inverse $x \mapsto (\alpha_i^ x \alpha_j)_{ij}$. It restricts to group isomorphisms $\mathrm{GL}_r(L) \cong L^\times$ for all $r \geq 1$. In particular $M_r(L) \cong L$ for every $r \geq 1$.*

Proof. The map $\Theta_{\mathcal{C}}$ is k -linear, sends the identity matrix to $\sum_i \alpha_i \alpha_i^* = 1$ by Lemma 5.1, and is multiplicative:

$$\Theta_{\mathcal{C}}(a) \Theta_{\mathcal{C}}(b) = \sum_{i,j,l,m} \alpha_i a_{ij} \alpha_j^* \alpha_l b_{lm} \alpha_m^* = \sum_{i,m} \alpha_i \left(\sum_j a_{ij} b_{jm} \right) \alpha_m^* = \Theta_{\mathcal{C}}(ab),$$

using $\alpha_j^* \alpha_l = \delta_{jl}$. The displayed inverse is two-sided by the same identities: $\alpha_i^* \Theta_C(a) \alpha_j = a_{ij}$, and $\sum_{i,j} \alpha_i (\alpha_i^* x \alpha_j) \alpha_j^* = 1 \cdot x \cdot 1 = x$. An algebra isomorphism restricts to an isomorphism of unit groups, and $\mathrm{GL}_r(L) = M_r(L)^\times$. $\square$

For every $r \geq 2$ there is a canonical choice, the *left comb*

$$(23) \quad \mathcal{C}_r = (s_1, s_0 s_1, s_0^2 s_1, \dots, s_0^{r-2} s_1, s_0^{r-1}),$$

the leaf set of the tree in which every left child is subdivided; the ordering lists the leaves by depth with the leftmost branch last.

5.2. Elementary groups over rings. Let A be a unital ring. Recall $x_{ij}(a) = I_r + a E_{ij}$ for $i \neq j$, the relations $x_{ij}(a) x_{ij}(b) = x_{ij}(a + b)$, and the Steinberg commutator relation

$$[x_{il}(a), x_{lj}(b)] = x_{ij}(ab) \quad (i, j, l \text{ distinct}),$$

where $[g, h] = ghg^{-1}h^{-1}$.

Lemma 5.3 (Whitehead commutator lemma). *Let A be a unital ring and $a, b \in A^\times$. Put*

$$w(a) = x_{12}(a) x_{21}(-a^{-1}) x_{12}(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}.$$

Then

$$\mathrm{diag}(a, a^{-1}) = w(a) w(-1) \in \mathrm{EL}_2(A),$$

and

$$\mathrm{diag}(aba^{-1}b^{-1}, 1) = \mathrm{diag}(a, a^{-1}) \mathrm{diag}(b, b^{-1}) \mathrm{diag}((ba)^{-1}, ba) \in \mathrm{EL}_2(A).$$

Consequently, for every $m \geq 2$, every product of commutators of units $c \in [A^\times, A^\times]$, and every diagonal position, the matrix with c at that position and 1 elsewhere on the diagonal lies in $\mathrm{EL}_m(A)$.

Proof. First, $x_{12}(a) x_{21}(-a^{-1}) = \begin{pmatrix} 1 - aa^{-1} & a \\ -a^{-1} & 1 \end{pmatrix} = \begin{pmatrix} 0 & a \\ -a^{-1} & 1 \end{pmatrix}$, and multiplying by $x_{12}(a)$ on the right adds a times the first column to the second, giving $w(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}$. Then

$$w(a) w(-1) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

The displayed product of three diagonal matrices has first entry $ab(ba)^{-1} = aba^{-1}b^{-1}$ and second entry $a^{-1}b^{-1}(ba) = 1$. For the last statement, embed $\mathrm{EL}_2(A) \hookrightarrow \mathrm{EL}_m(A)$ in any chosen pair of coordinates and multiply the resulting matrices for the commutator factors of c . $\square$

Lemma 5.4. *Let A be a finitely generated unital ring and $r \geq 3$. Then $\mathrm{EL}_r(A)$ is finitely generated: if A is generated as a unital ring by the finite set T , then $\mathrm{EL}_r(A)$ is generated by the finitely many matrices $x_{ij}(1)$ and $x_{ij}(t)$, $t \in T$, $i \neq j$.*

Proof. Let E be the subgroup generated by the listed matrices, and let $A_0 \subseteq A$ be the set of a with $x_{ij}(a) \in E$ for all $i \neq j$. By the addition relation, A_0 is an additive subgroup containing 1 and T ; by the Steinberg relation with a third index $l \notin \{i, j\}$ (here $r \geq 3$ is used), $a, b \in A_0$ implies $ab \in A_0$, since $x_{ij}(ab) = [x_{il}(a), x_{lj}(b)]$ and $x_{il}(a), x_{lj}(b) \in E$. Hence A_0 is a unital subring containing T , so $A_0 = A$ and $E = \text{EL}_r(A)$. $\square$

Theorem 5.5 (Ershov–Jaikin–Zapirain [9]). *Let A be a finitely generated associative unital ring and $r \geq 3$. Then $\text{EL}_r(A)$ has Kazhdan’s property (T).*

The theorem in [9] is stated for the universal finitely generated rings $\mathbb{Z}\langle x_1, \dots, x_d \rangle$; the general case follows since EL_r of a quotient ring is a quotient group of EL_r of the ring, elementary generators mapping to elementary generators, and property (T) passes to quotients.

5.3. Characteristic-two block factorizations. The following two identities, valid in characteristic two, are the engine of the explicit spine.

Lemma 5.6. *Let A be a unital ring of characteristic 2 containing a unital copy of $L_k(1, 2)$ with k of characteristic 2. Then, in $M_2(A)$:*

- (a) $x_{12}(s_1) x_{21}(t_1) x_{12}(s_1) = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix};$
- (b) $x_{21}(1 + t_0) x_{12}(1) x_{21}(1 + s_0) x_{12}(t_0) = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix}.$

Proof. (a) Since $1 + p_1 = p_0$ in characteristic 2,

$$\begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} 1 + s_1 t_1 & s_1 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix},$$

and then

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & p_0 s_1 + s_1 \\ t_1 & t_1 s_1 + 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix},$$

using $p_0 s_1 = s_0(t_0 s_1) = 0$ and $t_1 s_1 + 1 = 1 + 1 = 0$.

(b) Compute the product of the first two factors:

$$\begin{pmatrix} 1 & 0 \\ 1 + t_0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 + t_0 & t_0 \end{pmatrix}$$

(using $1 + t_0 + 1 = t_0$ in characteristic 2). Multiplying by the third factor,

$$\begin{pmatrix} 1 & 1 \\ 1 + t_0 & t_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 + s_0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 1 + t_0 + t_0 + t_0 s_0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix},$$

where the (1, 1)-entry is $1 + 1 + s_0 = s_0$ and the (2, 1)-entry is $(1 + t_0) + t_0(1 + s_0) = 1 + t_0 + t_0 + t_0 s_0 = 1 + 1 = 0$, using $t_0 s_0 = 1$. Finally,

$$\begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} 1 & t_0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & s_0 t_0 + 1 \\ 0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix},$$

since $s_0 t_0 + 1 = p_0 + 1 = p_1 = s_1 t_1$ in characteristic 2. $\square$

5.4. Thompson's group V in a Leavitt corner. Thompson's group V is the group of right-continuous bijections of the boundary $\{0, 1\}^{\mathbb{N}}$ given by *tree tables*: a pair of ordered complete leaf sets $(\beta_1, \dots, \beta_\ell)$, $(\alpha_1, \dots, \alpha_\ell)$ of common length ℓ , acting by the prefix substitutions $\beta_i x \mapsto \alpha_i x$. Two tables define the same element exactly when they have a common refinement, refinement being simultaneous subdivision of a source leaf and its target leaf. The group V is infinite, simple, and finitely presented [11, 6]; it is in fact of type F_∞ , and 2-generated [5].

Proposition 5.7 (Tree-table embedding). *For $g \in V$ given by a table $((\beta_i), (\alpha_i))$, put*

$$(24) \quad U_g = \sum_{i=1}^{\ell} \alpha_i \beta_i^* \in L.$$

Then U_g is well defined (independent of the table), $U_g U_h = U_{gh}$, $U_1 = 1$, each U_g is a unit with $U_g^{-1} = U_{g^{-1}}$, and $g \mapsto U_g$ is an injective homomorphism $V \hookrightarrow L^\times$.

Proof. Refining a table replaces a summand $\alpha\beta^*$ by $\alpha 0 (\beta 0)^* + \alpha 1 (\beta 1)^* = \alpha(s_0 t_0 + s_1 t_1)\beta^* = \alpha\beta^*$; since any two tables of the same element admit a common refinement, U_g is well defined. For multiplicativity, choose tables of g and h such that the source leaf set of g equals the target leaf set of h (refine both accordingly); then

$$U_g U_h = \sum_{i,j} \alpha_i \beta_i^* \gamma_j \delta_j^* = \sum_i \alpha_i \delta_i^* = U_{gh},$$

using $\beta_i^* \gamma_j = \delta_{ij}$ for the common leaf set, and the composed table represents gh . Also $U_1 = \sum_i \alpha_i \alpha_i^* = 1$ by Lemma 5.1, so each U_g is a unit with inverse $U_{g^{-1}}$. For injectivity, suppose $U_g = 1$ with a table $((\beta_i), (\alpha_i))$. Multiplying $\sum_i \alpha_i \beta_i^* = 1$ on the right by β_j gives $\alpha_j = \beta_j$ for each j —the products $\beta_i^* \beta_j = \delta_{ij}$ collapse the sum—so g is the identity substitution. This is the standard realization of the Higman–Thompson groups inside Leavitt algebras and their Cuntz-algebra completions; compare [16, 17]. $\square$

Lemma 5.8 (Corner embedding). *Let A be a unital ring containing a unital copy of $L = L_k(1, 2)$. For $g \in V$ put*

$$\rho(g) = p_0 + s_1 U_g t_1 \in A.$$

Then $\rho : V \rightarrow A^\times$ is an injective group homomorphism, and for all $g \in V$:

$$t_1 \rho(g) s_1 = U_g, \quad \rho(g) s_0 = s_0, \quad t_0 \rho(g) = t_0.$$

Proof. All computations take place in the subring generated by the Leavitt family, using $t_1 s_0 = 0$, $t_0 s_1 = 0$, $t_1 s_1 = t_0 s_0 = 1$, $p_0 s_0 = s_0$, $t_0 p_0 = t_0$, $p_0 s_1 = 0$, $t_1 p_0 = 0$. Multiplicativity:

$$\begin{aligned} \rho(g) \rho(h) &= (p_0 + s_1 U_g t_1)(p_0 + s_1 U_h t_1) \\ &= p_0^2 + p_0 s_1 U_h t_1 + s_1 U_g t_1 p_0 + s_1 U_g t_1 s_1 U_h t_1 \end{aligned}$$

$$= p_0 + s_1 U_g U_h t_1 = \rho(gh),$$

by Proposition 5.7. Also $\rho(1) = p_0 + p_1 = 1$, so $\rho(g) \in A^\times$ with inverse $\rho(g^{-1})$. The three displayed identities are immediate: $t_1 \rho(g) s_1 = t_1 p_0 s_1 + t_1 s_1 U_g t_1 s_1 = U_g$; $\rho(g) s_0 = p_0 s_0 + s_1 U_g t_1 s_0 = s_0$; $t_0 \rho(g) = t_0 p_0 + t_0 s_1 U_g t_1 = t_0$. Injectivity: $\rho(g) = 1$ forces $U_g = t_1 \rho(g) s_1 = t_1 s_1 = 1$, hence $g = 1$ by Proposition 5.7. $\square$

Thus ρ realizes V inside the corner $p_0 + s_1(\cdot)t_1$, acting as the identity on the complementary corner $s_0(\cdot)t_0$: this complementarity is the algebraic seed of the centralizer relation $[K, J] = 1$.

Proposition 5.9. *Thompson's group V is finitely generated and is not LEF. Moreover V is perfect.*

Proof. V is finitely presented and infinite [11, 6] and even 2-generated [5]. If V were LEF, then being finitely presented it would be residually finite by Lemma 2.6; but V is infinite and simple, so its only finite quotient is trivial and it is not residually finite. Perfectness: $[V, V]$ is a nontrivial normal subgroup of the simple group V (it is nontrivial since V is nonabelian, containing for instance a copy of every finite symmetric group permuting the leaves of a fixed tree), hence $[V, V] = V$. $\square$

6. THE COMPRESSION SCHEME IN ADJACENT RANKS

This section presents the algebraic heart of the paper in its natural generality. All statements below are identities in matrix rings over an arbitrary ambient ring; the only hypotheses that are not ring identities are the two memberships isolated in Theorem 6.7, and they are discharged explicitly in Section 7 and K -theoretically in Section 8.

Throughout, A is an associative unital ring containing a unital copy of $L = L_k(1, 2)$ for some field k ; we identify s_0, s_1, t_0, t_1 with their images in A and retain the notation $p_0 = s_0 t_0$, $p_1 = s_1 t_1$. Fix $m \geq 2$ and put

$$n = m + 1, \quad G = \text{EL}_n(A), \quad \Gamma = \{\hat{\gamma} = \text{diag}(\gamma, 1) \mid \gamma \in \text{EL}_m(A)\} \leq G.$$

Indeed $\Gamma \leq G$: an elementary matrix in the first m coordinates is an elementary matrix of rank n . The group $\Gamma \cong \text{EL}_m(A)$ is infinite, since $a \mapsto x_{12}(a)$ is injective and $A \supseteq L$ is infinite-dimensional over k .

We use the left comb (23) with n leaves:

$$(25) \quad \alpha_i = s_0^{i-1} s_1 \quad (1 \leq i \leq m), \quad \alpha_n = s_0^m,$$

a complete leaf set, so that $\alpha_i^* \alpha_j = \delta_{ij}$ and $\sum_{i=1}^n \alpha_i \alpha_i^* = 1$ by Lemma 5.1.

6.1. The comb compressor. Define the column $b \in A^{m \times 1}$, the row $c \in A^{1 \times m}$, and the matrices $u, u' \in M_n(A)$:

$$(26) \quad \begin{aligned} b &= \begin{pmatrix} s_1 \alpha_1^* \\ \vdots \\ s_1 \alpha_m^* \end{pmatrix}, & c &= (\alpha_1 t_1 \quad \cdots \quad \alpha_m t_1), \\ u &= \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix}, & u' &= \begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix}. \end{aligned}$$

Lemma 6.1. *The following identities hold in A :*

$$bc = p_1 I_m, \quad cb = 1 - \alpha_n \alpha_n^*, \quad t_0 b = 0, \quad cs_0 = 0, \quad b\alpha_n = 0, \quad \alpha_n^* c = 0.$$

Consequently $uu' = u'u = I_n$; thus $u \in \text{GL}_n(A)$ with $u^{-1} = u'$.

Proof. Entrywise: $(bc)_{ij} = s_1 \alpha_i^* \alpha_j t_1 = \delta_{ij} s_1 t_1 = \delta_{ij} p_1$. Next, $cb = \sum_{i=1}^m \alpha_i t_1 s_1 \alpha_i^* = \sum_{i=1}^m \alpha_i \alpha_i^* = 1 - \alpha_n \alpha_n^*$ by Lemma 5.1. The entries of $t_0 b$ are $t_0 s_1 \alpha_i^* = 0$; those of cs_0 are $\alpha_i t_1 s_0 = 0$; those of $b\alpha_n$ are $s_1 \alpha_i^* \alpha_n = 0$ and of $\alpha_n^* c$ are $\alpha_n^* \alpha_i t_1 = 0$, by orthogonality of distinct leaves. Then

$$uu' = \begin{pmatrix} s_0 t_0 I_m + bc & b\alpha_n \\ \alpha_n^* c & \alpha_n^* \alpha_n \end{pmatrix} = \begin{pmatrix} (p_0 + p_1) I_m & 0 \\ 0 & 1 \end{pmatrix} = I_n,$$

and likewise

$$u'u = \begin{pmatrix} t_0 s_0 I_m & t_0 b \\ cs_0 & cb + \alpha_n \alpha_n^* \end{pmatrix} = I_n. \quad \square$$

Proposition 6.2 (Compression). *For every $\gamma \in \text{EL}_m(A)$,*

$$(27) \quad u \hat{\gamma} u^{-1} = p_1 \widehat{I_m + s_0 \gamma t_0},$$

and on elementary generators $u \hat{x}_{ij}(a) u^{-1} = \hat{x}_{ij}(s_0 a t_0)$ for $i \neq j \leq m$. Consequently

$$(28) \quad K := u \Gamma u^{-1} = \langle \hat{x}_{ij}(s_0 a t_0) : a \in A, i \neq j \leq m \rangle \leq \Gamma,$$

and conjugation by u is an isomorphism $\Gamma \xrightarrow{\cong} K$; in particular $K \cong \text{EL}_m(A)$.

Proof. Block multiplication using Lemma 6.1:

$$\begin{aligned} u \hat{\gamma} u^{-1} &= \begin{pmatrix} s_0 \gamma & b \\ 0 & \alpha_n^* \end{pmatrix} \begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix} = \begin{pmatrix} s_0 \gamma t_0 + bc & b\alpha_n \\ \alpha_n^* c & \alpha_n^* \alpha_n \end{pmatrix} \\ &= \begin{pmatrix} s_0 \gamma t_0 + p_1 I_m & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

For $\gamma = x_{ij}(a) = I_m + a E_{ij}$, the block is $p_1 I_m + s_0 t_0 I_m + s_0 a t_0 E_{ij} = I_m + s_0 a t_0 E_{ij} = x_{ij}(s_0 a t_0)$. Since Γ is generated by the $\hat{x}_{ij}(a)$ and conjugation is a group isomorphism onto its image, K is generated by the displayed elements, which lie in Γ . $\square$

The compressed copy K sits inside Γ with all matrix entries confined to the corner $s_0(\cdot)t_0 = p_0 A p_0$ of A ; the complementary corner $p_1 A p_1$ is untouched, and this is the room in which the centralizing subgroup J will live.

6.2. The involution z and generation. Let $z \in M_n(A)$ agree with the identity matrix outside the coordinates 1 and n , where it is

$$(29) \quad z|_{\{1,n\}} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}; \quad \text{that is,} \quad z = I_n + (p_0 - 1)E_{11} + s_1 E_{1n} + t_1 E_{n1} - E_{nn}.$$

Lemma 6.3. $z^2 = I_n$; in particular $z \in \text{GL}_n(A)$. Moreover z centralizes K .

Proof. On the coordinates $\{1, n\}$,

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}^2 = \begin{pmatrix} p_0^2 + s_1 t_1 & p_0 s_1 \\ t_1 p_0 & t_1 s_1 \end{pmatrix} = \begin{pmatrix} p_0 + p_1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$

using $p_0 s_1 = 0$, $t_1 p_0 = 0$, $t_1 s_1 = 1$; the remaining coordinates are fixed. For the centralizing statement, put $r = s_0 a t_0$ and note

$$p_0 r = r = r p_0, \quad t_1 r = 0, \quad r s_1 = 0.$$

Let $i \neq j \leq m$. If $1 \notin \{i, j\}$, then z acts as the identity on the coordinates i, j and $z \hat{x}_{ij}(r) z = \hat{x}_{ij}(r)$. If $i = 1$: the matrix $r E_{1j}$ satisfies $z(r E_{1j}) z = (p_0 r E_{1j} + t_1 r E_{nj}) z = r E_{1j} z = r E_{1j}$, since the j -th column of z is e_j for $2 \leq j \leq m$. If $j = 1$: $z(r E_{i1}) z = r E_{i1} z = r p_0 E_{i1} + r s_1 E_{in} = r E_{i1}$. In each case $z \hat{x}_{ij}(r) z^{-1} = \hat{x}_{ij}(r)$, and these elements generate K by (28). $\square$

Define the second compressor

$$(30) \quad v = zu.$$

Then, by Proposition 6.2 and Lemma 6.3,

$$(31) \quad v \Gamma v^{-1} = z(u \Gamma u^{-1}) z^{-1} = z K z^{-1} = K = u \Gamma u^{-1} :$$

the two compressors have one common image. The quotient $vu^{-1} = z$ is precisely what generates the missing last row and column:

Proposition 6.4 (Generation). *The subgroup of $\text{GL}_n(A)$ generated by Γ and z contains $\text{EL}_n(A)$. Consequently, if u and z lie in $\text{EL}_n(A)$, then*

$$\text{EL}_n(A) = \langle \Gamma, z \rangle = \langle \Gamma, u, v \rangle.$$

Proof. For $2 \leq j \leq m$ and $a \in A$, we claim

$$z \hat{x}_{1j}(s_1 a) z = x_{nj}(a), \quad z \hat{x}_{j1}(a t_1) z = x_{jn}(a).$$

For the first: $z(s_1 a E_{1j}) z = (p_0 s_1 a E_{1j} + t_1 s_1 a E_{nj}) z = a E_{nj} z = a E_{nj}$, using $p_0 s_1 = 0$, $t_1 s_1 = 1$, and that the j -th column of z is e_j . For the second: $z(a t_1 E_{j1}) z = a t_1 E_{j1} z = a t_1 p_0 E_{j1} + a t_1 s_1 E_{jn} = a E_{jn}$, using $t_1 p_0 = 0$, $t_1 s_1 = 1$, and that the j -th row of z is $e_j^\top$. Thus $\langle \Gamma, z \rangle$ contains $x_{nj}(a)$ and $x_{jn}(a)$ for all $2 \leq j \leq m$ and $a \in A$, as well as all $\hat{x}_{ij}(a) = x_{ij}(a)$ with $i \neq j \leq m$. The remaining elementary matrices are supplied by the Steinberg relation, using $m \geq 2$ so that the coordinate $2 \notin \{1, n\}$ exists:

$$x_{1n}(a) = [x_{12}(a), x_{2n}(1)], \quad x_{n1}(a) = [x_{n2}(a), x_{21}(1)].$$

Hence $\langle \Gamma, z \rangle \supseteq \text{EL}_n(A)$. If moreover $u, z \in \text{EL}_n(A)$, then $\langle \Gamma, z \rangle \subseteq \text{EL}_n(A)$, giving the first equality; and since $z = vu^{-1}$, also $\langle \Gamma, u, v \rangle \supseteq \langle \Gamma, z \rangle = \text{EL}_n(A) \supseteq \langle \Gamma, u, v \rangle$. $\square$

6.3. The commuting corner copy of V . For $g \in V$, recall $\rho(g) = p_0 + s_1 U_g t_1 \in A^\times$ from Lemma 5.8, and define

$$j(g) = \text{diag}(\rho(g), 1, \dots, 1) \in \text{GL}_n(A), \quad J = \{j(g) \mid g \in V\}.$$

Proposition 6.5. *For every $g \in V$, the matrix $\text{diag}(\rho(g), 1, \dots, 1) \in \text{GL}_m(A)$ lies in $\text{EL}_m(A)$. Hence $J \leq \Gamma$, the map $j : V \rightarrow J$ is an isomorphism, and J is a finitely generated subgroup of Γ that is not LEF.*

Proof. By Proposition 5.9, V is perfect, so $g = \prod_i [a_i, b_i]$ for some $a_i, b_i \in V$. Since ρ is a homomorphism (Lemma 5.8), $\rho(g) = \prod_i [\rho(a_i), \rho(b_i)]$ is a product of commutators of units of A . The Whitehead commutator lemma (Lemma 5.3, with $m \geq 2$) places $\text{diag}(\rho(g), 1, \dots, 1)$ in $\text{EL}_m(A)$. Thus $j(g) \in \Gamma$. The map j is a homomorphism because ρ is, and it is injective because ρ is; so $J \cong V$, which is finitely generated and not LEF by Proposition 5.9. $\square$

No hypothesis on A beyond the presence of the Leavitt family enters Proposition 6.5: the corner copy of V is elementary *unconditionally*.

Proposition 6.6. $[K, J] = 1$ and $K \cap J = \{1\}$. Consequently $KJ \cong K \times J$.

Proof. Centralizing. Conjugating an elementary matrix by a diagonal matrix multiplies its coefficient: $j(g) \hat{x}_{il}(r) j(g)^{-1} = \hat{x}_{il}(\rho(g)^{[i=1]} r \rho(g^{-1})^{[l=1]})$, where the exponent $[i = 1]$ is 1 if $i = 1$ and 0 otherwise. For $r = s_0 a t_0$, Lemma 5.8 gives $\rho(g)r = \rho(g)s_0 a t_0 = s_0 a t_0 = r$ and $r\rho(g^{-1}) = s_0 a t_0 \rho(g^{-1}) = s_0 a t_0 = r$. Hence $j(g)$ commutes with every generator $\hat{x}_{il}(s_0 a t_0)$ of K from (28).

Trivial intersection. Suppose $j(g) \in K$. By (27), every element of K has the form $p_1 \widehat{I_m + s_0 \gamma t_0}$ with $\gamma \in \text{EL}_m(A)$; comparing $(1, 1)$ -entries, $\rho(g) = p_1 + s_0 \gamma_{11} t_0$. Sandwiching between t_1 and s_1 : on one side, $t_1(\rho(g) - 1)s_1 = t_1(s_1 U_g t_1 - p_1)s_1 = U_g - 1$, using $t_1 p_0 = 0$; on the other, $t_1(p_1 + s_0 \gamma_{11} t_0 - 1)s_1 = t_1(s_0 \gamma_{11} t_0 - p_0)s_1 = 0$, using $t_1 s_0 = 0$ and $t_1 p_0 = 0$. Hence $U_g = 1$, so $g = 1$ by Proposition 5.7, and $j(g) = 1$.

Since K and J commute and intersect trivially, the multiplication map $K \times J \rightarrow KJ$ is an isomorphism. $\square$

6.4. The Leavitt-corner theorem.

Theorem 6.7 (Leavitt-corner theorem). *Let A be an associative unital ring containing a unital copy of $L_k(1, 2)$ for some field k , let $m \geq 2$, and put $n = m + 1$. Let $u, z \in \text{GL}_n(A)$ be the matrices (26) and (29). Assume:*

- (i) $\text{EL}_m(A)$ and $\text{EL}_n(A)$ have property (T);
- (ii) $u, z \in \text{EL}_n(A)$; this holds, for instance, whenever $\text{GL}_n(A) = \text{EL}_n(A)$.

Then $\text{EL}_n(A)$ is not sofic.

Proof. Put $G = \text{EL}_n(A)$ and let $\Gamma, u, z, v = zu, K, J$ be as above. By hypothesis (i), G and $\Gamma \cong \text{EL}_m(A)$ are property-(T) groups, hence finitely generated, and Γ is infinite. By hypothesis (ii) and Proposition 6.4, $G = \langle \Gamma, u, v \rangle = \langle \Gamma, Q \rangle$ with $Q = \{u, v\} \subseteq G$. By Proposition 6.2 and (31),

$$u\Gamma u^{-1} = v\Gamma v^{-1} = K \leq \Gamma,$$

so both elements of Q conjugate Γ into itself. Fix $q_0 = u$, so $K = q_0\Gamma q_0^{-1}$. By Propositions 6.5 and 6.6, the subgroup $J \leq \Gamma$ is finitely generated with $[K, J] = 1$ and $K \cap J = \{1\}$, and $J \cong V$ is not LEF. If G were sofic, Corollary 4.2 (Theorem D) would force J to be LEF, a contradiction. $\square$

Corollary 6.8. *Let A be a finitely generated unital ring containing a unital copy of $L_k(1, 2)$, and let $m \geq 3, n = m + 1$. If $u, z \in \text{EL}_n(A)$ —for instance, if $\text{GL}_n(A) = \text{EL}_n(A)$ —then $\text{EL}_n(A)$ is an infinite, finitely generated, property-(T) group that is not sofic.*

Proof. Theorem 5.5 supplies hypothesis (i) of Theorem 6.7 for both ranks $m, n \geq 3$, so $\text{EL}_n(A)$ is not sofic and has property (T). It is finitely generated by Lemma 5.4, and infinite since it contains $J \cong V$. $\square$

Remark 6.9 (The trust surface of the scheme). Every statement of this section except hypotheses (i) and (ii) of Theorem 6.7 is an identity in $M_n(A)$, verifiable by finite computation from the two Leavitt relations. In particular, the corner copy of V is elementary with no hypothesis at all: Proposition 6.5 needs only the perfectness of V and the Whitehead lemma, so no generalized-elementary (GE) or K -theoretic assumption enters through J . The membership hypothesis (ii) concerns exactly two matrices, u and z . Section 7 discharges it by explicit elementary factorizations in characteristic two; Section 8 discharges it over every finite field via $K_1(L) = 0$ and the GE property.

7. THE EXPLICIT REALIZATION: NO K -THEORY

In this section $k = \mathbb{F}_2$ and $R = L_{\mathbb{F}_2}(1, 2)$; every statement and proof holds verbatim over any finite field of characteristic two. Fix $m \geq 3$, put $n = m + 1$, and let $G = \text{EL}_n(R)$, Γ, u, z, v, K, J be as in Section 6 with $A = R$. We now exhibit the two matrices u and z as explicit products of elementary matrices, discharging hypothesis (ii) of Theorem 6.7 with no K -theoretic input.

Proposition 7.1 (Explicit elementary factorization of the compressor). *For $1 \leq i \leq m$, put*

$$U_i = x_{ni}(1 + t_0) x_{in}(1) x_{ni}(1 + s_0) x_{in}(t_0) \in \text{EL}_n(R).$$

Then

$$u = U_m U_{m-1} \cdots U_1.$$

In particular $u \in \text{EL}_n(R)$, as a product of $4m$ elementary matrices.

Proof. By Lemma 5.6(b), applied in the pair of coordinates (i, n) , the matrix U_i agrees with the identity outside the coordinates i and n and equals $\begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix}$ on them; explicitly,

$$U_i = I_n + (s_0 + 1)E_{ii} + s_1 t_1 E_{in} + (t_0 + 1)E_{nn},$$

signs being immaterial in characteristic two. Put $P_k = U_k U_{k-1} \cdots U_1$. We claim, by induction on k , that

$$P_k = \sum_{i \leq k} (s_0 E_{ii} + s_1 t_1 t_0^{i-1} E_{in}) + \sum_{k < i \leq m} E_{ii} + t_0^k E_{nn}.$$

For $k = 1$ this is the displayed form of U_1 . For the inductive step, compute the rows of $P_{k+1} = U_{k+1} P_k$. For $i \notin \{k+1, n\}$, the i -th row of U_{k+1} is $e_i^\top$, so the i -th row of P_{k+1} equals that of P_k . The $(k+1)$ -st row of U_{k+1} is $s_0 e_{k+1}^\top + s_1 t_1 e_n^\top$; since the $(k+1)$ -st row of P_k is $e_{k+1}^\top$ and the n -th row of P_k is $t_0^k e_n^\top$, the $(k+1)$ -st row of P_{k+1} is $s_0 e_{k+1}^\top + s_1 t_1 t_0^k e_n^\top$. The n -th row of U_{k+1} is $t_0 e_n^\top$, so the n -th row of P_{k+1} is $t_0^{k+1} e_n^\top$. This is the claimed form for $k+1$.

Finally, for the left comb (25), $s_1 t_1 t_0^{i-1} = s_1 (s_0^{i-1} s_1)^* = s_1 \alpha_i^*$ for $1 \leq i \leq m$, and $t_0^m = \alpha_n^*$. Hence

$$P_m = \sum_{i=1}^m (s_0 E_{ii} + s_1 \alpha_i^* E_{in}) + \alpha_n^* E_{nn} = \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix} = u,$$

by (26). □

Lemma 7.2 (Explicit elementary factorization of the involution).

$$z = x_{1n}(s_1) x_{n1}(t_1) x_{1n}(s_1) \in \text{EL}_n(R).$$

Proof. Lemma 5.6(a), applied in the pair of coordinates $(1, n)$, produces exactly the matrix z of (29). □

Theorem 7.3 (The explicit spine). *For every $m \geq 3$, the group $\text{EL}_{m+1}(R)$ over $R = L_{\mathbb{F}_2}(1, 2)$ is infinite, finitely generated, has property (T), and is not sofic.*

Proof. The ring $R = L_{\mathbb{F}_2}(1, 2)$ is finitely generated (by s_0, s_1, t_0, t_1 over the prime field). Proposition 7.1 and Lemma 7.2 verify $u, z \in \text{EL}_{m+1}(R)$, and Corollary 6.8 applies. □

Proof of Theorem A. Theorem 7.3 gives every $m \geq 3$; the case $m = 3$ is the group $G = \text{EL}_4(R)$. Every invertible matrix entering the argument is exhibited as an explicit product of elementary matrices: u by Proposition 7.1, z (hence $v = zu$) by Lemma 7.2, and the diagonal matrices $j(g)$, $g \in V$, by the Whitehead factorization of Lemma 5.3 applied to a commutator expression of g in the perfect group V (Proposition 6.5). No K -theory is used. □

Remark 7.4 (Scope of the explicit spine). The factorizations of this section are identities in characteristic two and hold over any field of characteristic two; finiteness of the field is used only through finite generation of the ring in Theorem 5.5. Thus $\text{EL}_{m+1}(L_k(1, 2))$ is infinite, finitely generated, Kazhdan, and nonsofic, with all memberships explicit, for every finite field k of characteristic two and every $m \geq 3$. For rank $n = 3$ (that is, $m = 2$), hypothesis (i) of Theorem 6.7 for EL_2 is not supplied by Theorem 5.5; the rank-two and rank-three groups are reached in Section 8 through self-similarity instead.

8. RANK TWO AND THE FULL PANORAMA

The compression–centralizer mechanism closes already in rank two, over a smaller group Γ : the full diagonal unit group. Combined with matrix self-similarity and the vanishing of K_1 , this yields Theorem B.

8.1. The two-by-two realization.

Theorem 8.1 (Two-by-two Leavitt-corner theorem). *Let A be an associative unital ring containing a unital copy of $L_k(1, 2)$ for some field k . Suppose that $A^\times$ and $\text{EL}_2(A)$ have property (T) and that $\text{GL}_2(A) = \text{EL}_2(A)$. Then $\text{EL}_2(A)$ is not sofic.*

Proof. Put $G = \text{EL}_2(A) = \text{GL}_2(A)$ and

$$\Gamma = \{\text{diag}(a, 1) \mid a \in A^\times\} \leq G.$$

Then $\Gamma \cong A^\times$ is a property-(T) group, hence finitely generated, and it is infinite because $\rho : V \hookrightarrow A^\times$ is injective (Lemma 5.8).

The compressor. Put

$$u = \begin{pmatrix} s_0 & p_1 \\ 0 & t_0 \end{pmatrix}, \quad u' = \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix}.$$

Using $p_1 s_0 = s_1(t_1 s_0) = 0$, $t_0 p_1 = (t_0 s_1)t_1 = 0$, $t_0 s_0 = 1$, and $p_0 + p_1 = 1$:

$$uu' = \begin{pmatrix} s_0 t_0 + p_1 & p_1 s_0 \\ t_0 p_1 & t_0 s_0 \end{pmatrix} = I_2, \quad u'u = \begin{pmatrix} t_0 s_0 & t_0 p_1 \\ p_1 s_0 & p_1 + s_0 t_0 \end{pmatrix} = I_2,$$

so $u \in \text{GL}_2(A) = G$ with $u^{-1} = u'$. For $a \in A$, put

$$\kappa(a) = p_1 + s_0 a t_0.$$

Then $\kappa(1) = 1$ and $\kappa(a)\kappa(a') = p_1 + s_0 a a' t_0 = \kappa(a a')$ by the same identities, so κ restricts to an injective group homomorphism $A^\times \rightarrow A^\times$; and

$$u \text{diag}(a, 1) u^{-1} = \begin{pmatrix} s_0 a & p_1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix} = \begin{pmatrix} s_0 a t_0 + p_1 & p_1 s_0 \\ t_0 p_1 & t_0 s_0 \end{pmatrix} = \text{diag}(\kappa(a), 1).$$

Hence

$$K := u \Gamma u^{-1} = \{\text{diag}(\kappa(a), 1) \mid a \in A^\times\} \leq \Gamma.$$

The involution. Put

$$z = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}, \quad \text{so} \quad z^2 = \begin{pmatrix} p_0^2 + s_1 t_1 & p_0 s_1 \\ t_1 p_0 & t_1 s_1 \end{pmatrix} = I_2,$$

using $p_0 s_1 = 0$, $t_1 p_0 = 0$, $t_1 s_1 = 1$; thus $z \in \text{GL}_2(A) = G$ and $z^{-1} = z$. For any $c \in A$,

$$(32) \quad z \text{diag}(c, 1) z = \begin{pmatrix} p_0 c p_0 + p_1 & p_0 c s_1 \\ t_1 c p_0 & t_1 c s_1 \end{pmatrix}.$$

The element $\kappa(a)$ satisfies $p_0 \kappa(a) = \kappa(a) p_0 = s_0 a t_0$, $t_1 \kappa(a) = t_1$, and $\kappa(a) s_1 = s_1$; substituting into (32) gives $z \text{diag}(\kappa(a), 1) z = \text{diag}(s_0 a t_0 + p_1, 1) = \text{diag}(\kappa(a), 1)$. Hence z centralizes K , and $v := zu \in G$ satisfies $v \Gamma v^{-1} = z K z^{-1} = K$.

Generation. For any $a \in A$, the elements $1 + p_0 a t_1$ and $1 + s_1 a p_0$ are units of A , since $(p_0 a t_1)^2 = 0$ and $(s_1 a p_0)^2 = 0$ (using $t_1 p_0 = 0$ and $p_0 s_1 = 0$); so the diagonal matrices $\text{diag}(1 + p_0 a t_1, 1)$ and $\text{diag}(1 + s_1 a p_0, 1)$ lie in Γ . Substituting $c = 1 + p_0 a t_1$ and $c = 1 + s_1 a p_0$ into (32) and using $t_1 p_0 = 0$, $p_0 s_1 = 0$, $t_1 s_1 = 1$, $p_0^2 = p_0$:

$$(33) \quad z \text{diag}(1 + p_0 a t_1, 1) z = \begin{pmatrix} 1 & p_0 a \\ 0 & 1 \end{pmatrix} = x_{12}(p_0 a),$$

$$(34) \quad z \text{diag}(1 + s_1 a p_0, 1) z = \begin{pmatrix} 1 & 0 \\ a p_0 & 1 \end{pmatrix} = x_{21}(a p_0).$$

Next, the *corner swap*

$$d = s_1 t_0 + s_0 t_1 \in A^\times, \quad d^2 = s_1 t_1 + s_0 t_0 = 1,$$

gives $D = \text{diag}(d, 1) \in \Gamma$ with $D^{-1} = D$. Conjugation by D multiplies an upper coefficient on the left by d and a lower coefficient on the right by d ; since $d p_0 s_0 t_1 = d s_0 t_1 = p_1$ and $s_1 t_0 p_0 d = s_1 t_0 d = p_1$,

$$D x_{12}(p_0(s_0 t_1 a)) D = x_{12}(p_1 a), \quad D x_{21}((a s_1 t_0) p_0) D = x_{21}(a p_1),$$

with the inner matrices supplied by (33) and (34). Finally

$$x_{12}(a) = x_{12}(p_0 a) x_{12}(p_1 a), \quad x_{21}(a) = x_{21}(a p_0) x_{21}(a p_1),$$

so $\langle \Gamma, z \rangle$ contains every elementary matrix; hence $\langle \Gamma, z \rangle = G$, and since $z = v u^{-1}$,

$$G = \langle \Gamma, u, v \rangle, \quad u \Gamma u^{-1} = v \Gamma v^{-1} = K \leq \Gamma.$$

The commuting corner. Put $J = \{\text{diag}(\rho(g), 1) \mid g \in V\} \leq \Gamma$; membership is automatic here, since $\rho(g) \in A^\times$. By Lemma 5.8 and Proposition 5.9, $J \cong V$ is finitely generated and not LEF. The subgroups K and J commute:

$$\begin{aligned} \kappa(a) \rho(g) &= (p_1 + s_0 a t_0)(p_0 + s_1 U_g t_1) = s_1 U_g t_1 + s_0 a t_0 \\ &= (p_0 + s_1 U_g t_1)(p_1 + s_0 a t_0) = \rho(g) \kappa(a), \end{aligned}$$

using $p_1 s_1 = s_1$, $t_1 p_1 = t_1$, $t_0 p_0 = t_0$, $p_0 s_0 = s_0$, $t_0 s_1 = 0$, $t_1 s_0 = 0$. And $K \cap J$ is trivial: if $\kappa(a) = \rho(g)$, then sandwiching between t_0 and s_0 gives

$a = t_0\kappa(a)s_0 = t_0\rho(g)s_0 = 1$, so $\rho(g) = \kappa(1) = 1$ and $g = 1$ by injectivity of ρ .

Conclusion. The data $G, \Gamma, Q = \{u, v\}, q_0 = u, K, J$ satisfy all hypotheses of Corollary 4.2: G and Γ are finitely generated Kazhdan groups, Γ is infinite, $G = \langle \Gamma, Q \rangle$, each $q \in Q$ conjugates Γ into itself with $q_0\Gamma q_0^{-1} = K$, and $J \leq \Gamma$ is finitely generated with $[K, J] = 1$ and $K \cap J = \{1\}$. If G were sofic, $J \cong V$ would be LEF, contradicting Proposition 5.9. $\square$

8.2. K_1 vanishes and $\text{GL} = \text{EL}$.

Proposition 8.2. *Let k be any field and $L = L_k(1, 2)$. Then:*

- (i) $K_1(L) = 0$;
- (ii) *the unit group $L^\times$ is perfect*;
- (iii) $\text{GL}_r(L) = \text{EL}_r(L)$ for every $r \geq 2$.

Proof. The algebra L is the Leavitt path algebra of the rose with two petals, and it is purely infinite simple [1]. The K -theory of Leavitt path algebras [2] gives the exact sequence

$$K_1(k) \xrightarrow{1} K_1(k) \longrightarrow K_1(L) \longrightarrow K_0(k) \xrightarrow{1} K_0(k),$$

where both outer maps are multiplication by $2 - 1 = 1$, hence isomorphisms; therefore $K_1(L) = 0$, proving (i). For a purely infinite simple unital ring R , the natural map $R^\times \rightarrow K_1(R)$ induces an isomorphism $R^\times/[R^\times, R^\times] \cong K_1(R)$, and R is a GE-ring: every invertible matrix over R is a product of elementary matrices and a matrix $\text{diag}(a, 1, \dots, 1)$ with $a \in R^\times$ [3]. Applying this to $R = L$: (i) gives $L^\times = [L^\times, L^\times]$, proving (ii); and for $g \in \text{GL}_r(L)$, $r \geq 2$, write $g = e \cdot \text{diag}(a, 1, \dots, 1)$ with $e \in \text{EL}_r(L)$ and $a \in L^\times$. By (ii), a is a product of commutators of units, so $\text{diag}(a, 1, \dots, 1) \in \text{EL}_r(L)$ by the Whitehead commutator lemma (Lemma 5.3). Hence $g \in \text{EL}_r(L)$, proving (iii). $\square$

Remark 8.3. For an independent recent account of generation phenomena for unit groups and matrix groups over Leavitt algebras, consistent with Proposition 8.2 and with the self-similarity of Proposition 5.2, see [12].

8.3. Finite fields: all ranks and the unit group.

Theorem 8.4. *Let k be a finite field and $L = L_k(1, 2)$. Then $\text{EL}_2(L) = \text{GL}_2(L)$ is infinite, finitely generated, has property (T), and is not sofic.*

Proof. The ring L is finitely generated, by s_0, s_1, t_0, t_1 and the finitely many scalars. By Theorem 5.5 and Lemma 5.4, $\text{EL}_3(L)$ is a finitely generated Kazhdan group; by Proposition 8.2(iii), $\text{EL}_3(L) = \text{GL}_3(L)$; and by Proposition 5.2 with the left comb $\mathcal{C}_3$, $\text{GL}_3(L) \cong L^\times$. Hence $A^\times = L^\times$ is a finitely generated Kazhdan group. Similarly, $\text{GL}_2(L) = \text{EL}_2(L)$ by Proposition 8.2(iii), and $\text{GL}_2(L) \cong L^\times$ by Proposition 5.2 with $\mathcal{C}_2 = (s_1, s_0)$; so $\text{EL}_2(L)$ is Kazhdan as well. Theorem 8.1 with $A = L$ now shows that $\text{EL}_2(L)$ is not sofic. It is infinite, containing $\text{diag}(\rho(V), 1) \cong V$, and finitely generated, being Kazhdan. $\square$

Theorem 8.5 (Panorama). *Let k be a finite field and $L = L_k(1, 2)$. Then the groups*

$$L^\times = \mathrm{GL}_1(L), \quad \mathrm{GL}_r(L) \ (r \geq 1), \quad \mathrm{EL}_r(L) \ (r \geq 2)$$

are mutually isomorphic—explicitly, $\Theta_{\mathcal{C}} : \mathrm{GL}_r(L) \rightarrow L^\times$ is an isomorphism for every ordered complete leaf set $\mathcal{C}$ of size r , and $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ for $r \geq 2$ —and each is an infinite, finitely generated, property-(T) group that is not sofic.

Proof. The isomorphisms are Proposition 5.2, and the equalities $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ are Proposition 8.2(iii). Each of the listed groups is therefore isomorphic to $\mathrm{EL}_2(L)$, which is infinite, finitely generated, Kazhdan, and nonsofic by Theorem 8.4; all four properties are isomorphism-invariant. $\square$

Proof of Theorem B. Part (i) and part (ii) are exactly Theorem 8.5, with the explicit isomorphism (2) given by Proposition 5.2. $\square$

Remark 8.6 (Two routes to rank four). For a finite field of characteristic two and rank $m + 1 \geq 4$, the membership hypotheses are discharged twice over: explicitly, with no K -theory, in Section 7; and via Proposition 8.2 here. Theorem B in its full strength—all ranks down to two, all finite fields, and the unit group itself—uses the second route.

9. FINITELY PRESENTED NONSOFIC COVERS

Nonsficity of a finitely generated group is witnessed by a finite fragment of its multiplication table; imposing that fragment as a presentation produces finitely presented nonsofic covers. Soficity is not known to pass to quotients, and nothing of the sort is used: the cover is proved nonsofic directly.

Definition 9.1. Let G be a group, $F \subseteq G$ finite with $1 \in F$, and $\varepsilon > 0$. An (F, ε) -model of G is a map $\varphi : F \cup F \cdot F \rightarrow \mathrm{Sym}(Y)$, for some finite set Y , such that

$$(35) \quad d_H(\varphi(g)\varphi(h), \varphi(gh)) \leq \varepsilon \quad (g, h \in F),$$

$$(36) \quad d_H(\varphi(g), \varphi(h)) \geq 1 - \varepsilon \quad (g, h \in F, g \neq h).$$

Lemma 9.2 (Finite-table characterization). *A countable group G is sofic if and only if it admits an (F, ε) -model for every finite $F \ni 1$ and every $\varepsilon > 0$.*

Proof. If σ_n is a sofic approximation, then for fixed (F, ε) the restriction of σ_n to $F \cup F \cdot F$ is an (F, ε) -model for all large n : (35) is Definition 2.1(a), and (36) follows from (a) and (b) since $d_H(\sigma_n(g), \sigma_n(h)) \geq d_H(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(h)) - d_H(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(g)) = d_H(\sigma_n(gh^{-1}), \mathrm{id}) - o(1)$.

Conversely, exhaust G by finite symmetric sets $F_1 \subseteq F_2 \subseteq \dots$ containing 1, and let φ_n be an $(F_n, 1/n)$ -model on Y_n ; replacing Y_n by many disjoint copies of itself we may assume $|Y_n| \rightarrow \infty$. Define $\sigma_n(g) = \varphi_n(g)$ for $g \in F_n$ and $\sigma_n(g) = \mathrm{id}$ otherwise. Fix $g, h \in G$; for n large, $g, h, gh \in F_n$, and (35) gives Definition 2.1(a). For faithfulness, first note that $p_n = \varphi_n(1)$ satisfies

$d_H(p_n^2, p_n) \leq 1/n$ by (35) with $g = h = 1$, and $d_H(p_n^2, p_n) = d_H(p_n, \text{id})$ by right invariance of d_H ; so $d_H(\varphi_n(1), \text{id}) \leq 1/n$. Then for $g \neq 1$ and n large,

$$d_H(\sigma_n(g), \text{id}) \geq d_H(\varphi_n(g), \varphi_n(1)) - d_H(\varphi_n(1), \text{id}) \geq 1 - \frac{2}{n},$$

by (36). Hence σ_n is a sofic approximation. $\square$

Theorem 9.3 (Finite multiplication-table covers). *Let G be a finitely generated group that is not sofic. Then there is an infinite finitely presented group H that is not sofic, together with a surjection $H \twoheadrightarrow G$.*

Proof. Finite groups are sofic, so G is infinite. By Lemma 9.2 there are a finite set $F \subseteq G$ with $1 \in F$ and an $\varepsilon_0 > 0$ such that G admits no (F, ε_0) -model; enlarging F , we may assume F is symmetric and generates G . Define the finitely presented group

$$(37) \quad H = H_F = \langle x_g \ (g \in F \cup F \cdot F) \mid x_1 = 1, \ x_g x_h = x_{gh} \ (g, h \in F) \rangle.$$

The assignments $x_g \mapsto g$ satisfy the relations in G , and F generates G , so there is a surjection

$$(38) \quad \pi : H \twoheadrightarrow G, \quad \pi(x_g) = g.$$

In particular H is infinite, and $\pi(x_g) = g \neq h = \pi(x_h)$ shows $x_g \neq x_h$ in H for distinct $g, h \in F \cup F \cdot F$.

Suppose H were sofic, with sofic approximation $\sigma_n : H \rightarrow \text{Sym}(Y_n)$. Define $\varphi_n : F \cup F \cdot F \rightarrow \text{Sym}(Y_n)$ by $\varphi_n(g) = \sigma_n(x_g)$. For $g, h \in F$, the relation $x_g x_h = x_{gh}$ holds in H , so

$$d_H(\varphi_n(g)\varphi_n(h), \varphi_n(gh)) = d_H(\sigma_n(x_g)\sigma_n(x_h), \sigma_n(x_g x_h)) \rightarrow 0.$$

For distinct $g, h \in F$, the element $x_g x_h^{-1} \neq 1$ of H gives, as in the proof of Lemma 9.2,

$$d_H(\varphi_n(g), \varphi_n(h)) \geq d_H(\sigma_n(x_g x_h^{-1}), \text{id}) - o(1) \rightarrow 1.$$

Hence for n large, φ_n is an (F, ε_0) -model of G —note that (35) and (36) refer only to the multiplication table of F inside G , which is faithfully copied by the generators of H . This contradicts the choice of (F, ε_0) . Therefore H is not sofic. $\square$

Theorem 9.4 (Kazhdan covers). *Let G be a nonsofic group with property (T). Then there is an infinite finitely presented property-(T) group H that is not sofic, together with a surjection $H \twoheadrightarrow G$.*

Proof. Property-(T) groups are finitely generated, and G is infinite as before. By Shalom's theorem [19], G is a quotient of a finitely presented property-(T) group: fix a surjection $\theta : P \twoheadrightarrow G$ with P finitely presented and Kazhdan. Choose (F, ε_0) as in the proof of Theorem 9.3, and for each $g \in F \cup F \cdot F$ choose a lift $\tilde{g} \in P$ with $\theta(\tilde{g}) = g$, taking $\tilde{1} = 1$. Define

$$(39) \quad H = P / \langle\langle \tilde{g} \tilde{h} \widetilde{gh}^{-1} : g, h \in F \rangle\rangle.$$

Then H is finitely presented (finitely many new relators over the finitely presented P) and has property (T) (a quotient of P). Each relator lies in $\ker \theta$, since $\theta(\tilde{g}\tilde{h}\tilde{g}\tilde{h}^{-1}) = gh(gh)^{-1} = 1$; hence θ descends to a surjection $H \twoheadrightarrow G$, and H is infinite. Writing $y_g \in H$ for the image of $\tilde{g}$, we have $y_1 = 1$, $y_g y_h = y_{gh}$ for $g, h \in F$, and $y_g \neq y_h$ for distinct $g, h \in F$ (their images in G differ). If H were sofic, then $\varphi_n(g) = \sigma_n(y_g)$ would be an (F, ε_0) -model of G for large n , exactly as in the proof of Theorem 9.3, a contradiction. $\square$

Proof of Theorem C. The first two assertions are Theorems 9.3 and 9.4. For the final assertion, apply Theorem 9.4 to the infinite finitely generated property- (T) nonsolic group $L^\times$ of Theorem B, over any fixed finite field; the resulting H surjects onto $L^\times$, and composing with the inverses of the isomorphisms Θ_C of Theorem B(ii) exhibits a quotient of H isomorphic to each group listed there. $\square$

Remark 9.5 (Effectivity). The covers are effective in the logical sense: nonsolicity of G is equivalent, by Lemma 9.2, to the existence of a finite witness (F, ε_0) , and any witness yields H by (37) or (39). The proof establishes the existence of a witness without computing one, and no claim is made that $L^\times$ itself is finitely presented.

10. CONCLUDING REMARKS

Remark 10.1 (What the argument does and does not use). Theorem A is independent of $\text{GL}_r = \text{EL}_r$ and of $K_1(L) = 0$: every membership is an explicit factorization, and the analytic input is the four-line co-area estimate. The argument concerns permutation models only; it says nothing about hyperlinearity or the Connes embedding problem.

Remark 10.2 (Two compressors, one image). The pair $Q = \{u, v\}$ resolves a genuine tension. Generation of G requires elements that leave Γ ; compression requires elements that map Γ into itself. A single element cannot do both jobs in the criterion, because the criterion consumes the compressors twice: once through (4) to couple the ambient decomposition to the Γ -decomposition, and once through q_0 to seat K . Two compressors with one common image K , differing by the involution $z = vu^{-1}$, split the labor: K carries the compression, z carries the last row and column of EL_n .

Remark 10.3 (The median is canonical up to scale). The normalization $f = s/(s + m)$ of Section 4 pins at $1/2$, and the value $1/2$ is not a trace or a spectral constant: it is the fixed point of the exchange of source and target sizes, and it is a vertex-weighted median by construction. Median pinning is exactly the statement that a permutation compressing one expander decomposition into another must asymptotically preserve the size profile—the conservative core of the argument, with exact conservation (Lemma 3.2) converting a one-sided drift estimate into a two-sided one at no cost.

Remark 10.4 (What property (T) is used for). Property (T) enters three times, in three different roles: (a) through Theorem 2.7 applied to Γ , producing the source decomposition that is compressed; (b) through Theorem 2.7 applied to G , producing the ambient decomposition carrying the co-area inequality—and by Remark 4.3 this use is replaceable by any componentwise ℓ^1 Poincaré decomposition; (c) through the Kazhdan factor hypothesis of Theorem 2.8. The local Theorem 4.1 quantifies over the first two uses and leaves only the third to the black box.

Remark 10.5 (Scope). The Leavitt-corner theorem (Theorem 6.7) applies to an arbitrary unital ring containing a binary Leavitt family, whenever the two adjacent elementary groups are Kazhdan and the two matrices u, z are elementary; for finitely generated rings and $m \geq 3$ the Kazhdan hypothesis is automatic (Corollary 6.8). Finite fields are where all hypotheses are verified simultaneously (Theorem 8.5); characteristic two is where they are verified explicitly (Theorem 7.3); and rank two is where the realization is smallest (Theorem 8.1).

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